# FRC 2026 REBUILT -- Top-30 Robot Design Deep Dive

Sources: Chief Delphi, David's Design Server (frcdesign.org), team CAD and code releases, Behind the Bumpers, and Statbotics EPA.

## The 2026 design landscape

### The game

REBUILT is a fuel (6" foam ball) shooter: collect fuel and score it into your hub when active. Robots may hold unlimited fuel. The end-game tower climb existed but was low value: most top teams dropped or never built a climber, so design is about the fuel cycle: intake → storage/indexing → launch.

### The defining meta

The season split into two offensive families: (1) high-volume 'DUMPERS' -- huge hoppers that drive to the hub and empty fuel in bulk during scoring windows (1678 'Limestone', 254, 7769); and (2) 'SHOOTERS' -- flywheel or drum launchers, frequently TURRETED with vision auto-aim for shoot-on-the-move (4414, 6329, 1987, 2767, 5000, 4481). Several teams rebuilt mid-season from a passer/multi-wide shooter to a dumper (2910 Blitz→ReBlitz; 9470 triple→drum; 5687 'Big Dumpy').

### Indexing was where teams innovated

Moving fuel by geometry beat squeezing it. Spindexers ('car wash' dual-stack on 1796, zero-compression drum on 6329, cone-in-bowl on 9483), drum/revolver magazines, and roller-floor serializers (1678, 1768) dominated. Large flow-through hoppers (50-80 fuel) were common.

### Drivetrain & vision

Swerve is universal at this tier. Vision was a differentiator but was rarely published: confirmed stacks include 9470 (2× OV2311 global-shutter + Orange Pi 5 Pro, shoot-on-the-move derived from 6328), 1678 (Limelight 4 + CANrange), 3467 (custom Jetson + 4 cameras), 5000 (PhotonVision), 157 (PhotonVision on Orange Pi 5). Most teams left vision hardware unstated -- marked 'unconfirmed' here, meaning not published, not absent.

### How to use this document

Profiles are ordered by Statbotics EPA (season-cumulative scoring contribution). For each team we capture archetype, the four subsystems, vision, public CAD/design availability, and 'lessons for students'. The teams with the most reproducible documentation for a hands-on study are 1678, 4414, 1690, 7769, 6329, 6800, 3467, 9470, 5000, and 4481. Accuracy: subsystem specifics (motors, gear ratios, materials) are compiled from each team's public CAD, tech binder, reveal, and code, and may contain transcription or interpretation errors -- always confirm a detail against the team's own linked source before relying on it. Corrections from teams are applied verbatim (e.g., Team 5000's entry was corrected on 2026-07-06 at the team's request).

## Top-30 at a glance (by Statbotics EPA)

| # | Team | Robot | EPA | Rec | Archetype | Public CAD/Design |
|---|---|---|---|---|---|---|
| 1 | *** 4414 HighTide** | Rip Current | **357** | 70-2 | Single-turret flywheel shooter with 'dye-rotor' centering; hig | Tech binder |
| 2 | **254 The Cheesy Poofs** | Overload | **328** | 65-6 | 4-wide dumper with an articulating hood; ~25-28 BPS on fresh b | Team site |
| 3 | *** 7769 The CREW** | CHUNK | **312** | 70-5 | Wide single-stream dumper (chose dumper over turret) with a hi | CD thread |
| 4 | **1323 MadTown Robotics** | | **310** | 69-4 | Dye-rotor (drum) turret shooter + four-bar linkage intake [FRC | Stats/TBA |
| 5 | **2056 OP Robotics** | Desperate Measures | **302** | 63-10 | Mid-season rebuild: turret shooter, variable hood; one-motor r | CD thread |
| 6 | *** 1690 Orbit** | Kepler | **295** | 47-5 | 640deg turret shooter, shoot-on-the-move; roller-floor indexer | CAD release |

| # | Team | Robot | EPA | Rec | Archetype | Public CAD/Design |
|---|---|---|---|---|---|---|
| 7 | **27 Team RUSH** | | **295** | 83-6 | Wide 3-fuel steel-drum shooter fed by a roller floor + passive | Stats/TBA |
| 8 | *** 1114 Simbotics** | Simbot Tim | **288** | 62-7 | Wide drum dumper (deliberately no turret); large fuel capacity | Onshape CAD |
| 9 | **2481 Roboteers** | | **285** | 44-8 | Big dumper: fixed 4-wide shooter, 5-ball-wide touch-it-own-it | Stats/TBA |
| 10 | **5687 The Outliers** | Glenn / Big Dumpy | **283** | 62-12 | Fast drum-shooter that iterated into a dumper | Code (GitHub) |
| 11 | *** 2910 Jack in the Bot** | Blitz -> ReBlitz | **282** | 53-23 | Built a passer/3-wide shooter ('Blitz'), then rebuilt mid-seas | CAD release |
| 12 | **125 NUTRONs** | FLUX | **282** | 68-17 | Enlarged dye-rotor ('dish') shooter; trapezoidal drive base; d | Team site |
| 13 | *** 9470 Ctrl-Alt-Defeat** | | **279** | 60-13 | Triple shooter rebuilt mid-season into a drum shooter; 'CTRL+A | Tech binder |
| 14 | **7558 ALT-F4** | | **279** | 51-17 | Drum shooter with an advanced vision/simulation software progr | Team site |
| 15 | *** 1678 Citrus Circuits** | Limestone | **278** | 61-9 | Dumper / hooded shooter | CAD release |
| 16 | **6369 Mercenary Robotics** | | **274** | 77-9 | Rebuild from turret to wide drum shooter; slapdown kicker-ramp | Code (GitHub) |
| 17 | *** 6800 Valor** | Downpour | **268** | 62-10 | Elevator-launched fuel scorer with assisted-aim and bidirectio | Onshape CAD |
| 18 | *** 3467 Windham Windup** | | **266** | 75-10 | High-throughput dumper with auto-aim shooter backup; linear-sl | Onshape CAD |
| 19 | **9128 ITKAN Robotics** | Triple Threat | **265** | 60-7 | Drum shooter; cycloidal slapdown intake; duct-tape conveyor + | Video |
| 20 | **1796 RoboTigers** | Sprout | **263** | 48-7 | Fuel scorer: launcher fed from a 'Car Wash' spindexer; swerve | CD thread |
| 21 | **9483 Overcharge** | Enigma | **261** | 38-6 | Minimalist, low-motor-count spindexer scorer; 2 top rollers; s | Video |
| 22 | *** 6329 The Bucks' Wrath** | Roman I / Roman II | **260** | 61-15 | Wide aluminum drum shooter, net hopper; rebuilt from a turret | Onshape CAD |
| 23 | *** 2046 Bear Metal** | Bearricade MkII | **260** | 78-16 | Drum shooter, no turret; collector with CatTongue rollers; hop | Tech binder (Drive) |
| 24 | *** 1987 Broncobots** | Cyclone | **259** | 54-6 | 360deg turret dye-rotor shooter; linear deployable full-width | CAD release |
| 25 | *** 1768 Nashoba Robotics** | Nightshift | **259** | 70-17 | Drum shooter with a powered hood, no turret; fixed-azimuth ful | CAD release |
| 26 | **8044 Denham Venom** | Cat-5 | **257** | 50-9 | Dye-rotor shooter; v1 low 'bump'/under-trench, v2 a HighTide/D | DDS thread |
| 27 | *** 1706 Ratchet Rockers** | Mirage | **256** | 45-2 | Dual-turret shooter ('shooting so fast you think you're seeing | CD thread |
| 28 | **9496 LYNK** | Matterhorn | **254** | 70-13 | 2-wide drum 'over-bumper reverse' shooter (modeled on 254's 20 | CD thread |
| 29 | **5460 Strike Zone** | Helios | **252** | 57-11 | Drum shooter mounted on a turret (a 'turret dumper'), ~30deg t | CD thread |
| 30 | *** 581 Blazing Bulldogs** | Rubble | **251** | 47-19 | Dye-rotor turret shooter w/ 35-deg adjustable hood; pivoting f | Onshape CAD |

* = recommended study pick (high performance + a usable public CAD or design release).

# Team-by-team profiles

**#1 (study pick)** | **4414 HighTide** Rip Current | **EPA 357**
San Diego, CA | 70-2

## Single-turret flywheel shooter with 'dye-rotor' centering; high-BPS single stream

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|---|---|
| **SHOOTER**<br>4x Kraken X44 on a 3in flywheel; turret geared 55.7:1 with a flex-wheel cable-chain tensioner; low gearing keeps launch current low | **INTAKE**<br>Over-bumper/ground intake, 2 x Kraken X60 + 1 x X44 bottom bar; left/right sides decoupled for durability |
| **INDEXER / HOPPER**<br>'Dye rotor' optimized for high-BPS single-stream feeding; structural bumpers double as hopper walls and the shooter mount | **DRIVETRAIN**<br>25 x 32 in swerve, 2 custom modules from WCP Swerve X2 parts, 7.67:1; rear corners chamfered to grow hopper volume |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Vision auto-aim and shoot-on-the-move on the turret; controls-led, with precomputed trajectory maps. |
| **VISION**<br>Limelight always pointed at the hub (reads the hub's 4 AprilTags), accounting for ~95% of on-robot localization [BtB transcript] | **PUBLIC CAD / DESIGN**<br>Interactive tech binder (no downloadable CAD) |

**LESSONS FOR STUDENTS**

- Make structure multi-functional, as with the structural bumpers, saving weight and volume.
- The low flywheel gearing protects the whole electrical budget.
- The state-machine codebase is AI-assisted; read it with the binder as a documentation model.

**Tech binder:** https://2026.team4414.com/ · Tech binder · Reveal · Statbotics · Behind the Bumpers (full walkthrough)

| #2 | 254 The Cheesy Poofs Overload<br>San Jose, CA | EPA 328<br>65-6 |
|---|---|---|

## 4-wide dumper with an articulating hood; ~25-28 BPS on fresh balls [BtB transcript]

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|---|---|
| **SHOOTER**<br>4-wide dumper shooter with an articulating hood and a 'triangle of compression': each ball contacts the main drum, the bottom feeder roller and the top hood roller at once -> zero bobble, full spin-up, consistent shots. Only ~1.25in clearance per side, so a double-sided belt drives the main drum (outer) and the back hood rollers (inner) for the reversing effect. Backing feeder swapped poly -> CatTongue (poly slipped); shooter motors nested behind [BtB transcript] | **INTAKE**<br>Most-iterated subsystem. Rack-and-pinion deploy: racks went from 3/8in polycarb w/ titanium teeth (kept breaking) to two layers of 1/4in SRP + one 1/8in titanium (unbreakable). A bottom 3-roller rips balls up titanium kicker plates/bars (aluminum bent), tuned for low compression without dead spots; a 2nd roller keeps stored hopper balls from blocking incoming ones; dual Kraken X60. (Used GoPros inside to find jams; SRPP chosen to spread load around bolt holes, ~30% lighter, impact-resistant) [BtB transcript] |
| **INDEXER / HOPPER**<br>'Bumper hopper': a full 8in polycarb wall extending up from the bumpers (more width, fewer parts) + a carbon-fiber hoop/corners for rigidity; a net raises capacity from ~50 to ~75-80 balls. Hopper floor: WCP rubber-back timing belts (smoother than rollers). The floor's pivot is linked to the intake deploy ('trash compacting'): pulling the intake in while shooting lifts the floor so gravity empties the last balls; the intake fully stows under the floor [BtB transcript] | **DRIVETRAIN**<br>Swerve; deliberately does not shoot on the move (current limits) and instead optimizes for fast alignment + a defined pivot point [BtB transcript] |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Drive to the hub and empty fuel in bulk during scoring windows; Choreo 3-stage autos with a recovery path [BtB transcript] |
| **VISION**<br>Four cameras (their most ever) on one Jetson Orin Nano with GPU-accelerated AprilTag detection: two grayscale 120fps on the shooter side + two 60fps side cameras; multi-tag only (rejects single-tag), ~100+ fps. The Jetson solves and sends poses over UDP to the RIO, which filters by reprojection error + ambiguity. Heading control uses a predicted look-ahead pose + feed-forwards + an intelligent center-of-rotation (pivots about a corner for fastest alignment) [BtB transcript] | **PUBLIC CAD / DESIGN**<br>No public CAD (reveal video + build blog only) |

**LESSONS FOR STUDENTS**

- 254's build blog shows disciplined prototyping (backspin single-wheel vs dual flywheel) before committing.
- The team withheld CAD, so match video is your scouting source.
- Their strategic game-analysis method is as worth studying as the hardware.

**Team site:** https://www.team254.com/first/2026/ · 254 · Reveal · DDS robot-pics-2026 · Behind the Bumpers (full transcript)

#3 (study pick) **7769 The CREW** CHUNK — Michigan — **EPA 312** (70-5)

## Wide single-stream dumper (chose dumper over turret) with a high-capacity hopper

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| **SHOOTER**<br>Evolved from a two-chamber shooter to a 2.75in-wide shooter; praised for distance + accuracy. The hood surface is static rather than rollered. This removes the external belt that would otherwise run outside the hood plates and be damaged when the robot is pushed into the trench. With the hood up, the plates react against the trench instead of letting the robot be pushed underneath. Fuel does not jam at the 4 in stealth wheels because their surface speed is far higher than the feeding wheels. After a partial rebuild between the week 2 and week 4 events, powering both the top and bottom indexing rollers pushed fuel into the chamber too fast and jammed it. The fix was to fix the lower rollers so they cannot rotate and run the top rollers powered against a static lower surface; that configuration ran the rest of the season. | **INTAKE**<br>Linear slapdown intake; deploy arms cut from polycarbonate rack + 98A TPU printed chain tensioner; dynamic current limit (45A deploy -> 5A hold) for compliance. Each side is driven independently. Per side there are four 1/8 in polycarbonate plates: two with a flat bottom to slide on and two cut for the WCP 10dp racks, giving 1/2 in of slide width. The same 10dp aluminium gears ran all season; the racks were replaced once. The rack is fixed to the intake with heat-set inserts rather than threading into the polycarbonate, which survived repeated full-speed rams. The TPU idler is not strictly required: exact centre-to-centre spacing ran without it for the first two events, and it was added to keep the two sides in sync. |
| **INDEXER / HOPPER**<br>High capacity: ~70 fuel max, 50-60 in play; in/out throughput was their edge. The vectored wheels are print-in-place, adapted from a 2910 model with the roller-to-shaft clearance opened up so the rollers spin reliably and the wheel slips onto hex, plus a 1/8 in boss on one side so wheels stack without a spacer. Printed in PETG; one set lasted the whole season because they are protected inside the frame. The belt floor uses stock 105T x 15 mm timing belts. | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>The robot has a climber; the robot name came from a part on it that was mis-named in CAD. | **AUTONOMOUS**<br>Drive-and-empty dumper cycle. |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>Public CAD + tech slides (Onshape) |

**LESSONS FOR STUDENTS**

- Polycarb intake teeth sheared every 2-3 matches at champs; they ran ~16 spares. Choose intake materials for fatigue.
- Software current-limit tuning can buy mechanical compliance without extra hardware.
- A stated design rule, applied every year: build the simplest robot that meets the objectives the team decided are needed to be competitive. The static hood is that rule applied to one part.
- The jam fix went the opposite way to intuition: less drive, not more power. Before adding power to a jam, check if fuel is arriving too fast.

**CD thread:** https://www.chiefdelphi.com/t/frc-7769-cad-tech-slides-chunk/521008 · CAD + slides · Behind the Bumpers

#4 **1323 MadTown Robotics** — Madera, CA — **EPA 310** (69-4)

## Dye-rotor (drum) turret shooter + four-bar linkage intake [FRC StratAnalyst video, secondary source]

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|---|---|
| **SHOOTER**<br>Dye-rotor (drum) turret [FRC StratAnalyst video] | **INTAKE**<br>Four-bar linkage ground intake [FRC StratAnalyst video] |
| **INDEXER / HOPPER**<br>Unconfirmed | **DRIVETRAIN**<br>Swerve (assumed) |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>No public CAD/design release |

**LESSONS FOR STUDENTS**

- World Champion (20-1 at Champs) with no public documentation: design knowledge is not always shareable.
- When a team publishes nothing, use TBA match video as the primary scouting source.
- A candidate to contact directly rather than study from public CAD.

**Stats/TBA:** https://www.thebluealliance.com/team/1323/2026 · Team site · TBA · FRC StratAnalyst top-10 video

## #5 2056 OP Robotics Desperate Measures

Stoney Creek, ON

**EPA 302**

63-10

### Mid-season rebuild: turret shooter, variable hood; one-motor rack-extended intake; roller-floor hopper; MK5n swerve

**SHOOTER**

Turret on an 8 in ID contact bearing; turret gear is laser-cut 1/4 in 6061-T6. Rotation 1x Kraken X44 at 8:40 and 10:100, +/- 185 deg with an energy chain, homed on a magnetic prox switch. Hood 1x Kraken X44 at 8:40 and 10:155, 34 deg of travel, release angle 73.5 to 39.5 deg from horizontal, continuously variable. Flywheel 2x Kraken X60 at 28:34 3MGT2 driving a 4 in Fairlane wheel pressed onto a 1-1/4 in aluminium tube. Backspin kicker 1x Kraken X44 at 32:32 3MGT2 on a dual 2 in stealth wheel. Shot vector is computed kinematically with time-of-flight compensation for shooting on the move [technical binder]. The earlier Behind the Bumpers walkthrough describes the flywheel run at one fixed RPM [BtB transcript]

**INTAKE**

Linear-extending two-roller intake, polycarbonate side arms, 2 x 1 bash bar. Rollers 1x Kraken X60 on two 2 in silicone-coated polycarbonate rollers at 12:30 5MHTD. Extension 1x Kraken X44 driving a rack at 8:40 20DP with a 12-tooth 8DP pinion, using CTRE Dynamic Motion Magic for separate extend and retract speeds and a low stator current limit so the intake gives like a spring on impact [technical binder]. The front section is sprung down with surgical tubing not shown in the CAD and retracts passively against the bumper; the axle is 3/4 in solid aluminium shaft with 3/8 in bolts in each end; the mount is 1/4 in plate backed by a second 1/4 in plate on the sheet-metal frame [CD thread 525183]

**INDEXER / HOPPER**

Active roller floor of 1-1/2 in silicone-coated polycarbonate rollers, 1x Kraken X60 at 14-36 20DP and 18-25 5MHTD, with a CANrange aimed into the hopper to manage fuel level. Singulator is a dual rubber-coated timing-belt drive: 2x Kraken X44 on the feed conveyors at 12:36 5MHTD plus 1x Kraken X44 on a vertical kicker roller that sorts fuel collisions. Shooter feed is a dual 2 in stealth wheel on 1x Kraken X60 at 16:28 20DP [technical binder]. The Behind the Bumpers walkthrough adds a net hopper of about 62 fuel repurposed from prior years and a bare-Lexan slip roller before the conveyor as an anti-jam [BtB transcript]

**DRIVETRAIN**

Swerve Drive Specialties MK5n modules on Kraken motors at the R1 ratio, 15 ft/s top speed, SDS spike grip wheels. Chassis is 2 x 1 x 1/8 in aluminium tubing with a 1/16 in aluminium baseplate, 25.5 x 29.5 in maximum size [technical binder]. All non-formed parts are 6061 and all formed sheet metal is 5052 [CD thread 525183]

**ENDGAME**

The binder mentions only Haas-machined climber plates, in the fabrication section.

**AUTONOMOUS**

A PID drive-to-point algorithm, with a series of points forming a path and the robot transitioning between points without fully slowing down. Shift timing is automated: the robot reads FMS game data and match time, with a manual override, and decides when to shoot to maximise the offensive shift.

**VISION**

Three fixed Limelight 4 cameras tracking AprilTags. Localization is Limelight MegaTag2 pose estimation fused with wheel odometry under dynamic confidence weighting. Polygon zone localization drives turret targeting and avoids trench-hood collisions [technical binder]. The Limelight brackets are printed PA-CF rather than TPU; three failed over the season, one a complete shear along the layer lines [CD thread 525183]

**PUBLIC CAD / DESIGN**

Technical binder + two CAD drawing packages (PDF; no 3D model)

**LESSONS FOR STUDENTS**

- Compare the abandoned V1 'Desperate Times' with the competition robot 'Desperate Measures' side by side.
- Per the binder, 'Desperate Times' peaked near 14 BPS. That was not much better than hoppered single-stream shooters and was worse on average throughput.
- The low stator current limit is the spring, replacing a mechanical shock absorber.
- The team attributes the one-motor intake throughput to gearing, low compression through good geometry and high current limits, not more motors.
- The team says the camera brackets should have been TPU.
- The team notes the drawing package is less complete than usual because parts outsourced to Fabworks were made from STEP files, bypassing drawings.

**CD thread:** https://www.chiefdelphi.com/t/team-2056-op-robotics-2026-technical-binder-release/525183 · Technical Binder release (CD) · OPR26-B-2056 Technical Binder (PDF) · CAD Drawing Package - Desperate Measures (PDF) · CAD Drawing Package - V1 Desperate Times (PDF) · Season Recap · TBA · Behind the Bumpers (read full transcript)

| #6 (study pick) | **1690 Orbit** Kepler<br>Israel | **EPA 295**<br>47-5 |
|---|---|---|

## 640deg turret shooter, shoot-on-the-move; roller-floor indexer; ~60-ball hopper; low-profile trench-bot [BtB transcript]

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|---|---|
| **SHOOTER**<br>Compact 640deg turret (360+ per side): X60 drive through a low gearbox (low CG), self-made gears on an 8in bearing, ball path offset toward the back to tuck into the corner. Shooter tuned for slight/no spin (most consistent stream, lowest energy loss): back rollers belted to a custom main shooter wheel; motors on an SLS winch pulled tight to the wheel + hood motors on the plate side via bevel gears to minimize diameter; custom 3D-printed energy chain + 14ga wires + a raised 'neck' so the chain wraps tight enough for 640deg [BtB transcript] | **INTAKE**<br>Durable/collapsible/swappable wide intake: two silicone-coated rollers on a Kraken X60 + a static bottom kicker roller; deliberately no slapdown (poor impact collapse); instead an in-house rack-and-pinion deploy (polycarb racks + aluminum/brass gears); 25x50mm front extrusion + a polycarb/aluminum side sandwich; the whole U-frame lifts off via 3 screws per side for fast swaps [BtB transcript] |
| **INDEXER / HOPPER**<br>3 stages: (1) roller floor of 11 carbon-fiber silicone rollers on an X60 (2:1), roller-to-roller via 3mm GT2 belts, a slight reduction to ease the 4-wide -> 3.5-wide jam point, plus two un-coated rollers to cut friction; (2) two 30mm 5mm-HTD vertical belts on two independent X44 (~8 m/s) over a 1mm PTFE sheet (~10mm compression) + a vertical kicker roller (repurposed from their 2024 Doppler shooter) on an X44 to clear jams; (3) an active 'C' shooter ramp (3 rows of 2.25in flex wheels + 2 rows of 2in stealth wheels) on an X60 -> 8 m/s with zero spin into the turret [BtB transcript] | **DRIVETRAIN**<br>29.5x24.5in (wide side = intake); bumpers attach to the hopper walls (thin polycarb + a net for ~60-ball capacity). Mid-season swap from a custom MK5-based module to stock SDS MK5 modules after the defense bot pushed them around: the MK5's wide spike treads dig into the carpet vs their thin nitro treads; reinforced/integrated top plates. 'Defense mode' X-locks the swerve [BtB transcript] |
| **ENDGAME**<br>Low-profile 'trench-runner' geometry trades climb access for faster, safer collection; climbing is not a focus. | **AUTONOMOUS**<br>Shoot-on-the-move is one of the four stated design pillars. |
| **VISION**<br>A camera on the turret on an Orange Pi running PhotonVision (aimed at the hub's AprilTags) + two rear cameras for blind spots when the turret cable wraps or a camera drops; tilt + bump detection lets it shoot while tilted; an off-robot shoot-on-the-move simulation (gravity, air resistance, Magnus) precomputes trajectory polynomials the robot evaluates live [BtB transcript] | **PUBLIC CAD / DESIGN**<br>Full public CAD (Onshape + x_t) |

**LESSONS FOR STUDENTS**

- First full year on Onshape: a case study in CAD-tool migration and clean release practice.
- A full-floor roller intake maximizes pickup area and tolerates sloppy approaches.

**CAD release:** https://www.chiefdelphi.com/t/frc-orbit-1690-2026-robot-cad-release/520673 · CAD release · Reveal 'Kepler' · Video · Behind the Bumpers (full transcript)

#7

## 27 Team RUSH

Clarkston, MI

**EPA 295**

83-6

### Wide 3-fuel steel-drum shooter fed by a roller floor + passive hopper; high-throughput dumper; 113 lb [BtB transcript]

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|---|---|
| **SHOOTER**<br>One main steel 3in drum run by 4x Kraken X60 + two 1in hood rollers on the same X60s (spin ~2x faster to minimize backspin); 1x X44 rack-and-pinion hood; mostly plastic SRP + carbon-fiber structure for weight. Variable compression (aluminum rods ~5/8in, CF stability rods ~3/8in, final ~1/2in) to fit a retrofit shooter compactly; the feeder accelerates fuel before the shooter, so the shooter only adds the final boost [BtB transcript] | **INTAKE**<br>Main roller on 2x Kraken X44 (2.5:1) + a free-spinning carbon-fiber top roller + 3 passive inner tubes; built for quick fixes (it's at the robot's fighting face); the hopper is driven passively off the intake [BtB transcript] |
| **INDEXER / HOPPER**<br>Roller floor + indexer on 2x Kraken X60; floor = 1in polycarb rollers (5 with CatTongue tape) funneling to the 3-wide shooter via dead-axle 3D-printed guides; geared 1:1 in front, ~17:19 at the back to speed balls up; a free-spinning roller prevents jams (replaced a belted one). Iterated from belt/'sushi'/chain to all-polycarb rollers [BtB transcript] | **DRIVETRAIN**<br>Swerve; all electronics mounted on the bottom (frees space for the roller floor + shooter); the electrical cover attaches to the bumpers so its weight doesn't count against inspection; 3D-printed component mounts [BtB transcript] |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Main Limelight 4 (heavily weighted, accepts tags mainly from the rear given the shooting orientation) + left and right Limelights fused as one entity into the pose estimate for odometry during feeding / when the back can't see [BtB transcript] | **PUBLIC CAD / DESIGN**<br>No public CAD/design release |

**LESSONS FOR STUDENTS**

- A Hall of Fame team and Michigan State champion with no public release: execution and consistency can outweigh documentation.
- Michigan's competitive density (27, 7769, 5460...) is itself a design driver: strong regions force strong robots.

**Stats/TBA:** https://www.thebluealliance.com/team/27/2026 · Reveal thread · Video · TBA · Behind the Bumpers (full transcript)

#8 (study pick) **1114 Simbotics** Simbot Tim — Ontario, Canada — **EPA 288** — 62-7

## Wide drum dumper (deliberately no turret); large fuel capacity [BtB transcript]

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|---|---|
| **SHOOTER**<br>Big belt-driven drum, an X44 + 3 more motors; hood uses only ~8deg of travel. They chose a drum (more versatile) over a turret (didn't suit their play style) and chose not to go under the trench [BtB transcript] | **INTAKE**<br>2x Kraken X60; balls hit a kicker bar + main roller; an 'agitating' intake with multiple states: it can vibrate fast to push balls into the indexer harder, or run gentler [BtB transcript] |
| **INDEXER / HOPPER**<br>Extending hopper -> a moving 'rolling floor' (two rollers) carries balls up to the drum [BtB transcript] | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>A climber was designed and appears in the CAD. It was added on March 1 and removed by March 19 for weight, and stayed off the rest of the season as the game developed. | **AUTONOMOUS**<br>All autonomous paths were built in the PathPlanner GUI, on full-field AprilTag localization. A simple PID-to-point with a timeout runs after crossing the bump to confirm the crossing before shooting starts. Autos are time-based: an auto that falls behind gives up and returns to the team zone, which the team says made pre-match auto coordination with alliance partners easier [code release]. |
| **VISION**<br>3x Limelight cameras with MegaTag2 + AprilTags, fused with odometry for pose estimation [BtB transcript] | **PUBLIC CAD / DESIGN**<br>CAD Release (Onshape) |

**LESSONS FOR STUDENTS**

- The active-vs-passive roller split is a low-cost fix for bridging.
- The intake was iterated over many versions with 1678: an example of cross-team prototyping focused on a single subsystem.
- Study it as a benchmark for overall packaging and mechanism integration.
- A designed mechanism has to earn its weight, as the climber shows.

**Onshape CAD:** https://cad.onshape.com/documents/7959d159451177786d52dd9a/w/535bc31b66b0cc442967f1d0/e/a263c1a625ed8cf2484ae82f · CAD Release (CD) · Code Release (CD) · Code (GitHub) · CAD (Onshape) · Statbotics 1114 · The Blue Alliance 1114

#9 **2481 Roboteers** — Tremont, IL — **EPA 285** — 44-8

## Big dumper: fixed 4-wide shooter, 5-ball-wide touch-it-own-it intake, 115-ball hopper (no turret) [BtB transcript]

| | |
|---|---|
| **SHOOTER**<br>4-wide fixed shooter, ~20 BPS; 4 accelerator rollers for accuracy; 3x Kraken X44 on the shooter + 2x X44 + 1x X60 on feed rollers; grip-tape (skateboard tape) for consistent ball spin (+~40% accuracy), a late-season PVC-pipe guide (+~10%); 2-position linear-actuator hood (replaced a pneumatic piston to cut weight), flywheel RPM varied for distance [BtB transcript] | **INTAKE**<br>5-ball-wide 'touch-and-take' intake: 3in roller wrapped in grip tape on 2x Kraken X60, SRPP side plates with 'Flintstone' wheels; ball compressed between roller and ramp; anti-jam = a 2nd motor + a plate to stop the roller stalling [BtB transcript] |
| **INDEXER / HOPPER**<br>115-ball, 5-wide hopper funneling to the 4-wide shooter through the accelerator rollers [BtB transcript] | **DRIVETRAIN**<br>Custom swerve (designed in the 2024 offseason) with large wheels and an adjustable ratio (12-32 ft/s); geared low this year for torque/low current under the heavy fuel load [BtB transcript] |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>No public release (GitHub stops at 2025) |

**LESSONS FOR STUDENTS**

- Strong record with a low public footprint: strong results needn't be open-sourced.
- Study via match video; consider contacting the team directly.

**Stats/TBA:** https://www.thebluealliance.com/team/2481/2026 · Team site · TBA · Behind the Bumpers (full transcript)

#10 **5687 The Outliers** Glenn / Big Dumpy — Portland, ME — **EPA 283** — 62-12

## Fast drum-shooter that iterated into a dumper

| | |
|---|---|
| **SHOOTER**<br>Near-full-width drum shooter (the team calls it 'quick'); high cycle rate prioritized | **INTAKE**<br>Over-bumper roller intake: 2in upper + 1.25in lower roller with ~1/2in compression (minimized to avoid bogging the motor) |
| **INDEXER / HOPPER**<br>CTRE/Kraken 'Feeder' subsystem feeding a calculated-shot ('ShotCalculator') shooter; 3-roller intake (top/bottom rollers + deployer) | **DRIVETRAIN**<br>Swerve (team has historically experimented with differential swerve) |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>3 cameras (front, left, right) on Limelight and PhotonVision; AprilTag pose estimation (MegaTag) | **PUBLIC CAD / DESIGN**<br>Code public (GitHub, C++); no CAD release |

**LESSONS FOR STUDENTS**

- They minimized roller compression by experiment.
- A separate 'LifeBot/Big Dumpy' repo signals a deliberate mid-season pivot to the dumper meta.
- The team moved from Java to C++ and reports an average 1.5 ms loop time with no loop overruns. A C++ codebase (less common than Java) is a study for advanced programmers.

**Code (GitHub):** https://github.com/frc5687 · 'Glenn' reveal · GitHub · TBA · Robot code (C++)

| #11 (study pick) | **2910 Jack in the Bot** Blitz -> ReBlitz<br>Mill Creek, WA | **EPA 282**<br>53-23 |
|---|---|---|

## Built a passer/3-wide shooter ('Blitz'), then rebuilt mid-season into a dumper ('ReBlitz')

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|---|---|
| **SHOOTER**<br>Blitz: 3-fuel-wide shooter, whole robot aims like a 3-wide turret, ~30 fuel/sec; ReBlitz dumper hit 30+ balls/sec. The hood is belt driven rather than chain driven because there was not room for chain at the gear ratio they wanted. The team notes a single belt or chain draws similar current, belt drag only matters where there are many of them, and chain behaves poorly at high speed. Hood plates use flathead screws because a button head would rub the hopper. Three rollers visible inside the shooter are aluminium supports that fasten the backing plates with zip ties; they began as feeder rollers and were removed after they worsened shot consistency and throughput. Between DCMP and Worlds the brass flywheel was remade with the 10T pulley integrated into it. | **INTAKE**<br>Dual intakes on Blitz. Powered roller floor feeder, taken deliberately from 1678 and 6328, and chained rather than belted: the team states chain has much lower resistance, lets the whole feeder and hopper run off one chain run, and is a contributing factor in reaching 34 BPS. The intake hardstop is the bumper itself; the gap visible in CAD is there for bumper fabric thickness, and on the DCMP version they had to lower that bumper edge about 1/16 in to let the intake reach the floor. |
| **INDEXER / HOPPER**<br>Blitz used a multilayered indexer + intentionally small hopper (a noted tradeoff); ReBlitz went to an open dumper path. Two changes between DCMP and Worlds are credited with passing 30 BPS: the feeder and hopper were geared together on 4 motors instead of running as separate powertrains, and a Teflon-taped carbon fibre board was added in front of the hopper rollers so that fuel filling the hopper hits the board and slides down into the shooter entry. | **DRIVETRAIN**<br>Swerve. Electronics sit on a brainpan behind a polycarbonate bellypan shield. On Blitz they were mounted underneath, so access meant flipping the robot and removing the shield; on Re-Blitz some moved on top of the brainpan for access. The baseplate uses a rectangular lightening pattern rather than a triangular one: the team states it saves more weight, and a belly or brain pan is not a structural member so it does not need the stiffness a triangular pattern buys. |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>The electrical system is a roboRIO, PDP, Pigeon and a CANivore, with the swerve CANcoders, Pigeon and CANranges on one splice. There is no mini power module and no Ethernet switch; Ethernet goes straight into the radio, and the team reports no radio heat problem. | **PUBLIC CAD / DESIGN**<br>Onshape CAD + tech binders for BOTH robots (CD release, 2026-06-07) |

**LESSONS FOR STUDENTS**

- The rebuild was run to a published schedule: after the first event an engineering realignment meeting concluded the original robot was too integrated for iteration to reach the target, then 5 days were allotted to design, 1.5 weeks to assembly and 1 week to bring-up and tuning.
- Gains between DCMP and Worlds came from all three of robot iteration, driver practice including match simulation and defense practice, and strategy work, not from hardware alone.

**CAD release:** https://www.chiefdelphi.com/t/2910-cad-and-tech-binder-release-2026/521705 · CAD + Tech Binder release · Blitz reveal · Behind the Bumpers

#12 **125 NUTRONs** FLUX — Boston, MA — **EPA 282** (68-17)

## Enlarged dye-rotor ('dish') shooter; trapezoidal drive base; dye rotor originated by the team (2017) [BtB transcript]

| | |
|---|---|
| **SHOOTER**<br>Enlarged dye-rotor 'dish' (sized up this year for throughput) fed by a live-shaft 3in carbon-fiber drum (CatTongue tape + silicone wrap) with a 3-standoff kick bar that maintains ball compression through the path [BtB transcript] | **INTAKE**<br>Four-bar linkage deploy, modular (whole intake pops off via 2 shoulder bolts per side); a stiffener bar for aggressive driving; the intake is carved out so the robot can sit under the trench at match start [BtB transcript] |
| **INDEXER / HOPPER**<br>Live-shaft 3in CF drum + 3-standoff kick bar feeding the dye-rotor dish [BtB transcript] | **DRIVETRAIN**<br>Trapezoidal base: wider at the front (bigger intake) and narrower at the rear (balls rarely reach the back); 1/4in side plates (no 2x1 main structure); 2 stock WCP Swerve X2S modules at the front + 2 custom-machined X2S at the rear, shaped to the dish radius [BtB transcript] |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>No CAD release located |

**LESSONS FOR STUDENTS**

- A distributed 5-school co-op is a model for sustaining strong performance across a large org.

**Team site:** https://www.nutrons.com/ · Reveal thread · Behind the Bumpers · TBA

#13 (study pick) **9470 Ctrl-Alt-Defeat** — Half Moon Bay, CA — **EPA 279** (60-13)

## Triple shooter rebuilt mid-season into a drum shooter; 'CTRL+ALT+DEL' deadline-driven scope control

| | |
|---|---|
| **SHOOTER**<br>Drum shooter (mid-season rebuild from a Week-1 triple shooter): open fuel path, three feeder rollers on two Kraken X60. | **INTAKE**<br>Reinforced beater-bar intake: a flippable 2x1 aluminum bar (flip it when bent), then two 0.25 in SRP leaf-spring plates per side with the intake tube as a hardstop and a detachable roller mount. |
| **INDEXER / HOPPER**<br>Hopper with COTS aluminum hubs + compliant floor wheels carried over from the triple-shooter robot. | **DRIVETRAIN**<br>Swerve; a CANivore drives 18-motor shoot-on-the-move bursts. |
| **ENDGAME**<br>Engineering hours went to scoring and intake robustness, not the climber. | **AUTONOMOUS**<br>Empirical shot map (measured, not modeled; first-shot +250 RPM compensation from data) with shoot-on-the-move. |
| **VISION**<br>Two OV2311 global-shutter cameras + Orange Pi 5 Pro coprocessors; SOTM / vision / current-limiting adapted from 6328 and the NINJAS; 4.6 V brownout floor so failures degrade gracefully. | **PUBLIC CAD / DESIGN**<br>Public tech binder + GitHub |

**LESSONS FOR STUDENTS**

- Their 'CTRL+ALT+DEL' rule ships the robot 1.5 weeks before every event: the deadline decides what to cut, copy, or skip (applied 3x: build, rebuild, reinforce).
- Reinforce the proven failure point instead of chasing optimization: robustness over speed for playoffs.
- Agentic coding tools for boilerplate let juniors plan instead of writing Java from scratch.

**Tech binder:** https://9470techbinder.vercel.app/ · Tech binder · Engineering process · Behind the Bumpers

## #14 7558 ALT-F4

North York, ON

**EPA 279**

51-17

### Drum shooter with an advanced vision/simulation software program; rebuilt after their 2nd event [BtB transcript]

| | |
|---|---|
| **SHOOTER**<br>3in aluminum drum with silicone tubing + two 1.5in silicone hood rollers; drum powered by 4x Kraken X60 geared 1:1; plus a custom laser-cut 1/2in-thick x 4in-wide steel flywheel [BtB transcript] | **INTAKE**<br>Linear intake running on HTD parts with 10DP gear-tooth racks meshing 10DP gears [BtB transcript] |
| **INDEXER / HOPPER**<br>Unconfirmed | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Advanced vision/simulation software program drives auto-aim; routines in code. |
| **VISION**<br>Multiple Limelights with a localization pipeline: per-camera outlier rejection -> 'smart' MegaTag1/MegaTag2 selection via hysteresis -> staged standard-deviation calc; logs everything to simulate the robot without owning a field [BtB transcript] | **PUBLIC CAD / DESIGN**<br>No public release located |

**LESSONS FOR STUDENTS**

- Their 2024 turret-on-swerve + efficient diverter suggests strong mechanism-aiming experience.
- Ontario Technology Division champions [BtB transcript]: study the reveal video.

**Team site:** https://www.team7558.com/ · Team site · YouTube · TBA · Behind the Bumpers (full transcript)

## #15 (study pick) 1678 Citrus Circuits Limestone

Davis, CA

**EPA 278**

61-9

### Dumper / hooded shooter

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|---|---|
| **SHOOTER**<br>Hooded flywheel fed from a full-width 3.5in aluminum drum (grip-taped); hood on an X44 jackshaft for a wide shot-angle range | **INTAKE**<br>Full-width slapdown ground intake: a ~6 lb aggressive over-bumper roller (Kraken X60 pivot, silicone roller + kicker bar) |
| **INDEXER / HOPPER**<br>Roller floor (dead-axle + flex-wheel rollers) -> ball tunnel/serializer; expanding hopper; pragmatic 'strategically placed wood' guides | **DRIVETRAIN**<br>Swerve, SDS MK5n, 27 x 27 in |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Drive-and-empty dumper. |
| **VISION**<br>Limelight 4 (single rear unit) + 2x CANrange sensors; custom localization + a driver dashboard with a 'shift' system | **PUBLIC CAD / DESIGN**<br>Full CAD + code (Onshape, GitHub) |

**LESSONS FOR STUDENTS**

- The reveal that 'confirmed the dumper meta': a community thread analyzed what Limestone implied about strategy.
- Software/driver UX is a force multiplier, as the custom driver dashboard shows.
- Strong teams still use simple, cheap fixes, such as the wood guides.
- Open CAD + code makes this a reproducible study target.

**CAD release:** https://www.chiefdelphi.com/t/1678-citrus-circuits-2026-cad-and-robot-code-release/521535 · Reveal: Limestone · Onshape CAD · Behind the Bumpers

#16 **6369 Mercenary Robotics**
McKinney, TX
**EPA 274**
77-9

## Rebuild from turret to wide drum shooter; slapdown kicker-ramp intake -> rolling floor; net hopper [BtB transcript]

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|---|---|
| **SHOOTER**<br>3in aluminum drum (only ~1/6in wide -> too little momentum, so they added a 1.5 lb WCP stainless flywheel at 1:3; stainless beat brass, which wore down); drum on 4x Kraken X60 at 1:1; shooter geared 60:1 on a single X44 (down from 400:1); transfer = two 2in + one 1.25in rollers tilted back to passively funnel; back rollers two 1.25in silicone; ~20 ft/s for 60-70 balls; all motors packaged under the drum. Chose drum over turret for higher throughput + simpler/more-reliable software [BtB transcript] | **INTAKE**<br>Slapdown intake (kept from robot 1) with a full kicker ramp (vs a kicker bar) for max fuel contact time + sustained compression (Citrus-style push into the indexer); 3 bars lift fuel high so the rolling floor can empty the hopper without much agitation [BtB transcript] |
| **INDEXER / HOPPER**<br>Rolling floor of custom 3D-printed stub rollers linked by belts (the intake jackshaft is hidden inside a roller to use the full width; one roller pivots with the intake); hopper extends exactly 12in + a net -> ~50-55 fuel un-expanded, up to ~78 at 30in with the net up (for bump-overfill cycles) [BtB transcript] | **DRIVETRAIN**<br>Swerve: swapped WCP X2 -> SDS MK5N (easy to replace), geared for speed + torque [BtB transcript] |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Three Limelight 4s (two front, one back) giving full-field localization in teleop + auto; one-button point-and-drive aiming with X-lock against defense; auto-aim auto-adjusts during autos [BtB transcript] | **PUBLIC CAD / DESIGN**<br>Code on GitHub; no confirmed CAD |

**LESSONS FOR STUDENTS**

**Code (GitHub):** https://github.com/mercs6369 · Behind the Bumpers · GitHub · DDS robot-pics-2026

#17 (study pick) **6800 Valor** Downpour
Austin, TX
**EPA 268**
62-10

## Elevator-launched fuel scorer with assisted-aim and bidirectional shooting

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|---|---|
| **SHOOTER**<br>Dual carbon-fiber arm launcher fed up a ~47in elevator (~0.3 s travel); can shoot in two directions. Because the chassis is square, spacers pull the shooter plate inboard of the nut strips so the shooter could be made 4 fuel wide within the frame; the team notes how close the pulleys sit to the edge of the robot. | **INTAKE**<br>Rotating intake synchronized with the arm for hand-off |
| **INDEXER / HOPPER**<br>Internal feeder routing fuel intake -> launcher; designed for fast maintenance access | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>A second robot, the Manor version of Downpour that the team calls Drizzle, has its own CAD branch in the same Onshape document. |
| **VISION**<br>Top-mounted camera for assisted scoring/alignment with a reverse-kinematic software model (published in their tech packet) | **PUBLIC CAD / DESIGN**<br>Public CAD (Onshape) |

**LESSONS FOR STUDENTS**

- One living CAD assembly iterated all season (branches per version) keeps design history in one place.
- Maintenance-driven layout (top electronics, bottom battery plate, ziptie battery-lead hooks): design for the pit, not just the field.
- Published a full Tech + Engineering-Ops binder: documentation discipline.

**Onshape CAD:** https://cad.onshape.com/documents/15cd41462dc24ff986a4e549 · Tech binder (PDF) · CAD release thread · Reveal · Behind the Bumpers

#18 (study pick) **3467 Windham Windup** — Windham, NH — **EPA 266** — 75-10

## High-throughput dumper with auto-aim shooter backup; linear-slide retracting intake; Thrifty Narrow swerve

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|---|---|
| **SHOOTER**<br>Primarily a bulk dumper (fill hopper, empty during on-shifts) with an auto-aim shot vs tower/trench | **INTAKE**<br>Linear-slide retracting intake with polycarbonate hopper walls; tested 1- vs 2-motor intake |
| **INDEXER / HOPPER**<br>Large hopper + indexer/tower rollers; tuned indexer floor angle and speed for throughput | **DRIVETRAIN**<br>Swerve on Thrifty Narrow modules (switched from MK4i), CANivore/CANcoder, Choreo paths |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Software/autonomous-led design; Choreo paths. |
| **VISION**<br>Custom NVIDIA Jetson + 4-camera setup; AprilTag localization | **PUBLIC CAD / DESIGN**<br>Full public CAD (Onshape) + Open Alliance blog |

**LESSONS FOR STUDENTS**

- Autonomous consistency beats peak score. The team was passed at alliance selection for inconsistent autos, then fixed odometry (lowered Choreo friction coeff, restored drivetrain current limits).
- New-carpet and CAN-terminator faults caused match losses; document environmental variables.
- Their Open Alliance blog is a model of transparent post-event self-critique.
- Deliberate 'simple week-1 bot, planned mid-season upgrade' scheduling.

**Onshape CAD:** https://cad.onshape.com/documents/7ef2c49c5bb48212a6b180ee · Open Alliance blog · CAD

#19 **9128 ITKAN Robotics** Triple Threat — Plano, TX — **EPA 265** — 60-7

## Drum shooter; cycloidal slapdown intake; duct-tape conveyor + transfer; passive 'hopper flopper' [BtB transcript]

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| **SHOOTER**<br>Drum shooter: 2x Kraken X60 on a ~13 lb flywheel mass (steel flywheels + wheels); bang-bang control (from their FTC shooter experience) to ramp fast, an operator pre-ramp button while crossing the bump, and a variable transfer speed (slows/speeds vs the flywheel velocity-vs-setpoint so each ball takes max energy); ~25-30 BPS [BtB transcript] | **INTAKE**<br>Flip-down slapdown intake on a cycloidal pivot drive (Kraken X60 -> cycloidals; chosen for ease of use + better ball compression than a linear intake); two rollers (3in hard wheels + a roller bar) on 2x Kraken X44; a pin-and-slot links the intake raise to pull the hopper back in (agitation) [BtB transcript] |
| **INDEXER / HOPPER**<br>7-roller conveyor with duct-tape flaps to fling balls into the transfer (the tape jumped BPS from ~18-20 to ~25-30); transfer = three 1in rollers + bottom stationary tubes for compression, rollers on slots for tuning. Passive 'hopper flopper': a flexible hopper that collapses to go under the trench and re-expands to store ~80 fuel [BtB transcript] | **DRIVETRAIN**<br>Swerve; built as a trench robot [BtB transcript] |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>A single Limelight facing the shooting direction; planned a 2nd but used software instead: an auto 'point-towards' the opposing hub to update odometry before crossing the bump (so it doesn't hit the trench) [BtB transcript] | **PUBLIC CAD / DESIGN**<br>No public CAD (reveal + Behind the Bumpers) |

**LESSONS FOR STUDENTS**

- The drum shooter trades flywheel spin-up for high burst rate, suited to empty-the-hopper volleys.
- Distinguish burst throughput from sustained throughput when specifying a shooter; they are different targets.
- Running two near-identical robots (9128/10340) multiplies build and test learning.

**Video:** https://www.youtube.com/watch?v=5HtLpv4AtlE · Reveal 'Triple Threat' · Behind the Bumpers

#20 **1796 RoboTigers** Sprout — EPA 263

Long Island City, NY — 48-7

## Fuel scorer: launcher fed from a 'Car Wash' spindexer; swerve

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|---|---|
| **SHOOTER**<br>Launcher fed from a spindexer | **INTAKE**<br>Intake pre-fills/'primes' the feeder while the outer spindexer stack turns slowly |
| **INDEXER / HOPPER**<br>'Car Wash' spindexer: two independently driven stacks, each on its own Kraken; outer stack spins slow during intake to evenly pack the hopper | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>No public CAD (photos on request) |

**LESSONS FOR STUDENTS**

- The slow spin during intake is a simple control trick for hopper packing.
- Other teams cite it as a top robot.

**CD thread:** https://www.chiefdelphi.com/t/robotigers-team-1796-presents-sprout-2026/516769 · Reveal 'Sprout'

#21 **9483 Overcharge** Enigma — EPA 261

Istanbul, Türkiye — 38-6

## Minimalist, low-motor-count spindexer scorer; 2 top rollers; swerve

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|---|---|
| **SHOOTER**<br>Launcher fed from a spindexer | **INTAKE**<br>Two top rollers + a bumper-level mechanism (deliberately simple, no obvious kickers/ramps) |
| **INDEXER / HOPPER**<br>Spindexer 'bowl' with a central cone wrapped in cat-tongue grip tape to fling fuel outward to the feeder | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>No public CAD (reveal video) |

**LESSONS FOR STUDENTS**

- The team has a reputation for 'doing things with fewer motors than anyone thinks possible'. Fewer actuators means fewer failure modes and lighter, cheaper robots.

**Video:** https://youtu.be/tOYFbRIJ8CA · Reveal 'Enigma'

| #22 (study pick) | **6329 The Bucks' Wrath** Roman I / Roman II<br>Bucksport, ME | **EPA 260**<br>61-15 |
|---|---|---|

## Wide aluminum drum shooter, net hopper; rebuilt from a turret + low-compression spindexer (Roman I) [BtB transcript]

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|---|---|
| **SHOOTER**<br>Roman II: wide aluminum drum (custom 3D-printed segments over an ordered tube, CatTongue-taped) + an adjustable hood; flywheel speed and hood angle scale with distance (further = faster + hood up); ~23 BPS (vs ~8 on the Roman I turret). A safety routine auto-lowers the hood entering the trench, with a bigger safety zone the faster it crosses [BtB transcript] | **INTAKE**<br>Custom silicone wheels (numbers printed) + an agitator linking all the rollers, CatTongue-taped for grip; a compact redesign over a wider v1 (a long bar that kept bending on hits was removed; flexible polycarb instead of Lexan to absorb impact); custom 3/8in-hex stub rollers instead of full-length shafts to save weight [BtB transcript] |
| **INDEXER / HOPPER**<br>A net hopper raised capacity from ~35 to ~60-70 fuel; rollers + agitator feed CatTongue-taped rollers up into the drum (the whole top net/cover flips up for easy access to the battery, PDH, and motors) [BtB transcript] | **DRIVETRAIN**<br>Swerve; built for serviceability (the top flips up to reach the battery / PDH / intake & swerve motors); prototyped on a wooden chassis to place electronics before final build [BtB transcript] |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>3 Limelights on AprilTags for pose: a third camera on the shooter added to the 2 side cameras for more accurate pose. One button locks onto the target or passing mode; an operator button auto-aligns when pose drops [BtB transcript] | **PUBLIC CAD / DESIGN**<br>Public CAD -- both versions (Onshape) |

**LESSONS FOR STUDENTS**

- Zero-compression indexing (move fuel by geometry/rotation, do not squeeze it) reduces jams and wear. Roman I spindexer: 20.75in outer, 6in center drum.
- A simple 'upkicker' (a static bar and side wheels) is a low-part-count hand-off from the spindexer to the turret.
- Both full CAD versions (Roman I and II) are released: compare them.

**Onshape CAD:** https://cad.onshape.com/documents/014716f840fdbd882c1d691f · Reveal 'Roman' · Behind the Bumpers · Roman II CAD

| #23 (study pick) | **2046 Bear Metal** Bearricade MkII<br>Maple Valley, WA | **EPA 260**<br>78-16 |
|---|---|---|

## Drum shooter, no turret; collector with CatTongue rollers; hopper-floor 'big dumper'; swerve

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|---|---|
| **SHOOTER**<br>4 lb, 4in-wide aluminum drum coated in CatTongue tape, ~3000 rpm; 24-25 BPS average (peaks 29-30), 4x Kraken X60; X44 hood with no rollers (rollers hurt accuracy); hood -10deg to 60deg (60deg = bump-to-bump pass; -10deg shoots backward against the hub under defense while locking the wheels) [BtB transcript] | **INTAKE**<br>Collector on 2x Kraken X60; two sets of CatTongue rollers + a 3rd 'pop-up' roller; stub axles for weight/durability; no front beater bar: 1/2in polycarb + aluminum plates at contact points for resilience without the weight; an upper TPU-mounted protector roller keeps balls in the hopper [BtB transcript] |
| **INDEXER / HOPPER**<br>Hopper extends 11.5in (kept under 12in so flex doesn't breach frame perimeter), double-fold polycarb top for rigidity + a net for capacity; ~70-80 fuel full, 50-60 while still fitting under the trench. Hopper bed = 11 rollers at a 21:20 ratio (back roller faster for grip), belts ~0.03in loose to cut current; floor pivots up for battery access. S-shaped (vs J-shaped) superstructure with a single 2-Kraken feed roller for throughput + less jamming + more hopper volume [BtB transcript] | **DRIVETRAIN**<br>Swerve (AdvantageKit-style logging of swerve states + odometry) [BtB transcript] |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>PathPlanner paths deployed as .path files, selected at the field by a 3-stage chooser, so routines can be recombined without redeploying code. |
| **VISION**<br>Vision pose updates fused with swerve odometry (logged); the 2025 code used PhotonVision + OV9782 AprilTag cameras [BtB transcript] | **PUBLIC CAD / DESIGN**<br>Bearricade Mk.II CAD (Google Drive) |

**LESSONS FOR STUDENTS**

- 'MkII' naming reflects a deliberate v1 to v2 redesign. Iterate on a known-good base rather than restarting.
- The team is known for fabrication quality and wire management. Clean execution scales to deep playoff runs.
- Pre-champs EPA peaked ~301: use it as a peak-performance benchmark.

**Tech binder (Drive):** https://drive.google.com/file/d/1heATIApMaUC-STLuVx7_C361IcmWv8U0/view?usp=sharing · CAD and Tech Binder release (CD) · Bearricade Mk.II CAD · Bearricade Mk.II Tech Binder · TBA · Behind the Bumpers

| #24 (study pick) | **1987 Broncobots** Cyclone<br>Lee's Summit, MO | **EPA 259**<br>54-6 |
|---|---|---|

## 360deg turret dye-rotor shooter; linear deployable full-width intake; 21.75in tall to clear the trench; WCP X2i swerve

**SHOOTER**
Turret shooter kept to 4.25in height so the dye-rotor hopper keeps its volume. Flywheel = 4in WCP 60A urethane wheel on a dead axle with a custom integrated pulley hub, driven by two Kraken X60 at 1.25:1; no flywheel mass was added, so RPM changes quickly for shoot-on-the-move. 'Floating hood' driven from the rear through a Thriftybot cycloidal gearbox on a Kraken X44 at 80.5:1 total, with a 10 lb constant-force spring that absorbs cycloidal backlash and a WCP through-bore encoder for true position; hood angle 0-90deg. Integrated 1in 45A hood rollers on a Kraken X60 at 28:20 reduce backspin. The floating hood was chosen over rack and pinion because a 4in rack limited hood angle. [Technical Book; CD release thread reply]

**INTAKE**
Linear deployable full-width intake that collects up to 5 fuel at once, sized for a 30in effective intake width to take the 5 outermost neutral-zone columns in auto. 2-roller system moves fuel front-to-back. Polycarbonate structure with aluminium hub covers, rigid-mounted so impact loads go into the chassis, and large washers to spread roller loads. Rollers: 1.125in silicone and cat-tongue-wrapped polycarbonate on a Kraken X60 at 1.5:1; deployment on a Kraken X60 at 3.6:1. [Technical Book]

**INDEXER / HOPPER**
Dye rotor, 22in diameter, mounted on its own plate for removal and service. 3D-printed diverter walls funnel fuel into the rotor; a wedge-shaped hook reduces the top footprint and allows reverse rotation to clear jams. Printed in Polymaker Fiberon PA612-ESD to prevent static buildup and reinforced with aluminium rods that carry turret load. Gearbox is coaxially driven with #25 chain on two X-contact bearings with fully enclosed bevel gears: rotation Kraken X60 at 20.67:1, feed wheels Kraken X60 at 1:1. Stated figures: 50 fuel hopper capacity, 8 fuel per rotor ring, 2.5 rotor rot/sec, 95 rot/sec interior wheels, 18 fuel/sec feed rate. Turret structure is sheet metal supported from the corner by a 1.5x1.5in tube that also carries the wires, reinforced with a carbon-fibre round tube; turret chain-driven by a Kraken X60 at 25.6:1 with a matched-reduction gearbox so the encoder reads absolute position. [Technical Book]

**DRIVETRAIN**
Four West Coast Products X2i swerve modules with molded tread. Custom 3/8in billet frame rails place the battery nearer the frame edge and fully support the dye-rotor centre column. An electronics brainpan carries most wiring on its underside, lowering the dye rotor while keeping access. 0.125in polycarbonate skid plate protects the electronics. Single-piece bumpers backed by aluminium sheet metal include integrated hopper walls. [Technical Book]

**ENDGAME**
Unconfirmed

**AUTONOMOUS**
B-Line autonomous routines dynamically recover from bumps and impacts from other robots. Intake speed scales with robot-relative velocity to conserve power.

**VISION**
Limelight 4 powered localization state with dynamic field zoning for automation; the Behind the Bumpers interview describes three Limelights for real-time pose. Shoot-on-the-move uses velocity control with augmented translational correction, gyro latency compensation, dynamic separation of fuel exit-velocity components for distance compensation through variable hood angles, and empirically measured fuel time-of-flight data for vector prediction. Real-time swerve wheel-slip detection corrects odometry. Code is Java command-based with a superstructure state machine and a Signal Fetcher that consolidates CAN interaction. [Technical Book; BtB transcript]

**PUBLIC CAD / DESIGN**
Onshape CAD + GitHub code + 14-page technical book

**LESSONS FOR STUDENTS**

- Turret wiring: the umbilical runs from below the brainpan, across a tensioner, and up an aluminium tube into the turret, so all harness management sits under the robot. The release thread lists what worked and what did not. A UHMWPE rope inside the harness takes tension off the wires. Braided sleeving failed at the redirects and was replaced with automotive fabric heat-shrink. Talc inside and Sailkote outside cut abrasion and friction. Fixed 3D-printed plus PTFE-tape redirects were used because pulleys on bearings would not meet the bend radius.
- Shooter height was set by the largest component, the flywheel. The added height also raised the turret structure enough to stop fuel jamming between the rotor hook and the structure.
- The cycloidal gearbox on the hood gave the reduction in a small package but had more backlash than expected.
- The technical book states measurable design targets before mechanisms, for example 15+ fuel/sec throughput, 10+ fuel/sec shooter flow and sub-1s spin-up.

**CAD release:** https://www.chiefdelphi.com/t/1987-broncobots-2026-cad-code-resources-release/524348 · Reveal 'Cyclone' · Behind the Bumpers · CAD, Code & Resources release (CD) · CAD (Onshape) · Robot code (GitHub) · Technical Book (PDF)

#25 (study pick) **1768 Nashoba Robotics** Nightshift — Bolton, MA — **EPA 259** (70-17)

## Drum shooter with a powered hood, no turret; fixed-azimuth full-width floor-loader; swerve

| | |
|---|---|
| **SHOOTER**<br>4x Kraken X60 geared ~1.35:1 onto a 4in dia 1/16" wall aluminum drum wrapped with CatTongue tape; two 1.375in hood rollers geared 1:1.4 off the main drum; ~3lb external steel flywheel belted 1:2.57 off the drum; hood pivot driven by a Kraken X44 on a 20DP rack (3D print sandwiched between two 1/16" aluminum laser-cut plates); separate entry roller on 2.25in squish wheels geared 2:1 off the roller-floor motors [tech slides] | **INTAKE**<br>4-bar: two 2in polycarbonate rollers on WCP silicone tubing, 2x Kraken X60 geared 2.2:1, plus two static 1/2 hex kicker bars; roller drive changed from belt to #25 chain to stop skipping; roller plates, side hopper plates and 6mm upper links in SRPP, with polycarbonate plates added in the transmission paths to hold tension against SRPP creep; 1/4" aluminum lower links on a dead axle give a consistent hard stop; deploy by one Kraken X60 at 90:1 (45:1 MaxPlanetary, then double #25 chain to a MaxSpline jackshaft under the roller floor and one #25 chain per side to TTB billet sprockets at the upper links); 1/8" wall aluminum bash bar reinforced with 1/8" wall steel u-channel [tech slides] |
| **INDEXER / HOPPER**<br>Roller floor on 3x Kraken X60 geared 1:1; three through-shafts of squish wheels feed the uptake, the remaining rollers 1.125in dia 1/32" wall aluminum with silicone tubing; WCP dead-axle stubs and custom plugs; belt distances shortened 0.030" to reduce current draw. Stationary hopper 3mm polycarbonate attached to the bumpers. Moving hopper rides the upper intake links and stays under trench height deployed or retracted: 3mm SRPP sides, 3mm polycarbonate back, braced by a carbon fiber tube with Formlabs-printed boots and free-rotating tubes that keep fuel out of the hopper/intake gap; stretchy net gives roughly 60 fuel (tech slides) or 65 fuel (reveal thread) [tech slides] | **DRIVETRAIN**<br>Swerve (confirmed 2026-10-13 by the dashboard owner) |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Autonomous Award winner at the New England District Championship. |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>CAD release (Onshape, both robots) + tech slides |

**LESSONS FOR STUDENTS**

- You do not need a turret to be competitive: the fixed-azimuth shooter plus a large hopper wins on throughput.
- One powered degree of freedom (the hood pivot) buys range without the cost of a turret.
- SRPP and dead-axle construction keep a wide superstructure light and stiff. SRPP was chosen where parts fractured.
- Budget current deliberately: shortening belt distances measurably reduced draw.
- A compliant net hopper trades rigidity for capacity.

**CAD release:** https://www.chiefdelphi.com/t/team-1768-2026-cad-release/525192 · CAD release (CD) · 'Nightshift' CAD (Onshape) · 'Nightshade' CAD (Onshape) · Technical Slides (PDF) · Reveal 'Nightshift' · Behind the Bumpers

#26 **8044 Denham Venom** Cat-5 — Denham Springs, LA — **EPA 257** (50-9)

## Dye-rotor shooter; v1 low 'bump'/under-trench, v2 a HighTide/Denham 'tide clone' [per 8044 team members on DDS]

| | |
|---|---|
| **SHOOTER**<br>Dye-rotor shooter, belt-driven (HighTide-style); v1 had belt skip only when jammed, v2 fixed it by tensioning the belt rather than adding 10DP gear stages [per 8044 on DDS; medium confidence] | **INTAKE**<br>Unconfirmed (2025 robot noted for a clean intake-to-handoff path; likely carryover) |
| **INDEXER / HOPPER**<br>Dye-rotor fed (rotor flings fuel to the launcher) [per DDS photos] | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>No public CAD (DDS robot-pics photos) |

**LESSONS FOR STUDENTS**

- Execution and driver practice can outrank publicity.
- Study their history of mechanism handoffs for reliable ball transfer.

**DDS thread:** https://discord.com/channels/1120162219502608426/1479491912305217637 · DDS robot-pics-2026 · TBA · DDS robot-pics-2026 thread (8044)

#27 (study pick) **1706 Ratchet Rockers** Mirage — Wentzville, MO — **EPA 256** (45-2)

## Dual-turret shooter ('shooting so fast you think you're seeing double') fed by a single spindexer [BtB transcript]

| | |
|---|---|
| **SHOOTER**<br>Two turrets, each with a front kicker + a back kicker; 6in aluminum flywheels (made by sponsor Custom Machine Works; tried steel, aluminum won); hoods rotate fully [BtB transcript] | **INTAKE**<br>Full-width four-bar intake, 2x Kraken X60 (one for pitch, one for the rollers); 4 silicone rollers (3 carbon-fiber + 1 polycarbonate, all wrapped in white silicone) for grip; the intake doubles as a hopper extension to minimize moving parts [BtB transcript] |
| **INDEXER / HOPPER**<br>A spindexer (re-iterated several times over the season) ramps fuel up into each shooter; kicker wheels fire the front + back kicker on each turret [BtB transcript] | **DRIVETRAIN**<br>Swerve (odometry fused with vision in the pose estimator) [BtB transcript] |
| **ENDGAME**<br>Telescoping-tube climber (Thrifty). | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>3x Limelight 4 + swerve odometry in the pose estimator; the Limelight hardware-manager Kalman filter rejects extraneous poses and only trusts MegaTag2 on good solutions; Limelights sit under each shooter + on the climber [BtB transcript] | **PUBLIC CAD / DESIGN**<br>Onshape CAD (champs config 'Mirage') + code on GitHub |

**LESSONS FOR STUDENTS**

- Consistency over flash: minimize failure modes and maximize cycle reliability.
- Watch their Behind the Bumpers for how a veteran team packages a clean, serviceable robot.

**CD thread:** https://www.chiefdelphi.com/t/522276 · CAD release (Mirage, champs) · Behind the Bumpers · Code (GitHub) · TBA

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#28 **9496 LYNK** Matterhorn — Spindale, NC — **EPA 254** (70-13)

## 2-wide drum 'over-bumper reverse' shooter (modeled on 254's 2017 'Misfire'); roller floor; swerve

| | |
|---|---|
| **SHOOTER**<br>2-wide drum shooter, fixed 34deg hood (polycarb + foam, ~0.675in compression); 4 motors on one shaft (SDS brass + TTB-0106 flywheels, ~5.2 lb); flywheel RPM scaled by distance | **INTAKE**<br>Roller floor with 'pulsing' (reverse-intake while shooting to empty the hopper) |
| **INDEXER / HOPPER**<br>Roller floor: indexer rollers 0.375in, top floor roller 0.5in; indexer/floor/shooter run 1:1 at 12V | **DRIVETRAIN**<br>MK5N-style swerve, ~26x26in, Spike grip wheels |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Single AprilTag camera for pose (primarily off the hub tags); a 2nd camera under consideration | **PUBLIC CAD / DESIGN**<br>Open Alliance build log + GitHub |

**LESSONS FOR STUDENTS**

- A young team (3rd year) shows the value of an Open Alliance build log: public iteration accelerates learning.
- Out-of-state district scheduling (VA, GA, SC) is a program-building lesson, not only a robot one.

**CD thread:** https://www.chiefdelphi.com/t/lynk-9496-build-log-2026/511797 · Open Alliance build log · Reveal 'Matterhorn'

#29 **5460 Strike Zone** Helios — Lapeer, MI — **EPA 252** — 57-11

## Drum shooter mounted on a turret (a 'turret dumper'), ~30deg total turret travel; swerve

| | |
|---|---|
| **SHOOTER**<br>Drum shooter on a ~30deg-travel turret | **INTAKE**<br>Unconfirmed |
| **INDEXER / HOPPER**<br>Unconfirmed | **DRIVETRAIN**<br>Swerve (historically SDS MK4i) |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>Release pending (strong tech-binder history) |

**LESSONS FOR STUDENTS**

- Strike Zone habitually publishes a full tech binder (2023, 2025): review those as a documentation template.
- A team known for 'accurate' shooters underscores investing in shot consistency over raw range.

**CD thread:** https://www.chiefdelphi.com/t/5460-strike-zone-2026-reveal/515689 · 'Helios' reveal · 2025 CAD + binder

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#30 (study pick) **581 Blazing Bulldogs** Rubble — San Jose, CA — **EPA 251** — 47-19

## Dye-rotor turret shooter w/ 35-deg adjustable hood; pivoting floor intake; WCP X2i/X2c swerve

| | |
|---|---|
| **SHOOTER**<br>Dye-rotor turret ('floating turret' de-conflicts rotor table vs turret). Drum: 4 in OD x 1/16 in-wall aluminum wrapped in Cat Tongue grip tape, powered by four Kraken X60s. Adjustable hood with 35 deg of rotation on a Kraken X44 at 1:120 through a zombie roller; two 2 in Cat Tongue hood rollers geared up 1.64:1 plus three 1.5 in silicone-sleeved rollers on two X60s (1:1.5). Chose the short dye-rotor turret over a large dumper after week-1 Alpha Bot testing hit ~15 BPS but lacked consistency. | **INTAKE**<br>Pivoting floor intake: pivots up/down for bellypan access and ball compaction; elastic X-pattern springing keeps constant tension so balls don't jam under the floor, and the front roller is lowered so balls can't cam under during the compaction sequence. Custom Onyx-printed pulleys with 1/16 in steel dowels to prevent layer shearing; installs onto the shooter tower with two bearing plates. |
| **INDEXER / HOPPER**<br>Floor rollers feed the shooter tower. A CANrange measures hopper fill and auto-runs the rollers until a retroreflective sensor in the tower trips - buffering ~8 extra fuel in the tower without ejecting prematurely. | **DRIVETRAIN**<br>28.5 x 27 in frame (1/8 in-wall tube, 5/8 in corner chamfers to stay under the 110 in perimeter). Two WCP X2i + two X2c swerve modules, X1 gearing with 10T pinions, Kraken X60 drive / X44 steer; custom 1/16 in bottom plates to drive over the Depot, intake-side module top plates cut down for a lower intake. |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Limelight-3 fuel-cluster-mapping autos with three midline lanes. |
| **VISION**<br>3x Limelight 4 (rear 'SHOOTER' + left + right), AprilTag MegaTag2; Limelight-only pipeline (no PhotonVision) | **PUBLIC CAD / DESIGN**<br>Full robot CAD (Onshape) + code (GitHub, team581/frc-2026) |

**LESSONS FOR STUDENTS**

- A quick archetype test is cheap insurance against a season-long mistake.
- The software power manager runs in a separate thread and re-allocates current limits across subsystems in real time (e.g. prioritize the shooter while scoring) to prevent brownouts over a full match.
- Defensive autonomy: if all cameras drop, the robot tells the drivers and falls back to pre-set shots. Dashboard toggles disable complex features mid-match.
- Turret wire management without e-chain: cable sleeving and surgical tubing for tension. It is low-cost and light.

**Onshape CAD:** https://cad.onshape.com/documents/766d37b113335d7a60cc2375/w/8b35fc383279c1954cec5b72/e/bd3188f47c23702bef9f1fc8 · CAD + Code release (CD) · Reveal (full Q&A) · Code (GitHub)

# Additional well-documented robots for study (beyond the EPA top-30)

These rank below the EPA top-30 but each has a strong public release (CAD, code, binder, or detailed reveal) and a clear design lesson. Ordered by EPA. ★ marks the recommended study picks.

**#4 (study pick)** — **2231 OnyxTronix** — Israel — **EPA 312** (38-7)

### 2x Kraken X60 turret shooter with a dynamic hood; ground fuel intake; 3-belt conveyor + trigger; buddy climb; swerve

| | |
|---|---|
| **SHOOTER**<br>Flywheel shooter powered by 2x Kraken X60 (1.25:1). A dynamic hood/arc is driven by a Kraken X44 (58:1). The shooter sits on a 2x Kraken X60 turret azimuth (4.347:1) with continuous polar-velocity tracking around the HUB. Stated throughput ~12 fuel/s. | **INTAKE**<br>Ground fuel intake. Rollers powered by 1x Kraken X60 (9.6:1); a 1x Kraken X44 (4.5:1) opens and closes the system linearly. The catalog lists full-width, multi-side, station and depot intake capability. |
| **INDEXER / HOPPER**<br>Conveyance in two parts. Conveyor: pulleys and timing belts driven by a Kraken X60 (3.125:1) plus a horizontal roller on the same motor (2.14:1); 3 conveyor belts keep ~3 fuel staged. Trigger: 3 synchronized rollers driven by a Kraken X60 (5:1), ~0.4 s intake-to-trigger. A passive, expandable 'cartridge' rides with the intake and holds up to ~50 fuel. | **DRIVETRAIN**<br>4x SDS MK5n swerve modules: Kraken X60 drive (R1 ratio 7.03:1) and Kraken X44 steer (approx 26.09:1). |
| **ENDGAME**<br>Buddy Climb capability and a reliable endgame climb are listed as season goals. | **AUTONOMOUS**<br>Won the Autonomous Award during the ISR district season. |
| **VISION**<br>Limelight 4 on the shooter/turret for pose-based hood-angle and flywheel-velocity control and continuous turret tracking around the HUB. The catalog describes a custom pose/state filter with dynamic covariance scaling and line-of-sight shot suppression near the climb structure. | **PUBLIC CAD / DESIGN**<br>Robot Catalog (PDF) |

**LESSONS FOR STUDENTS**

- No downloadable CAD source files were linked in the release thread, so the Robot Catalog is the study artifact.

**CD thread:** https://www.chiefdelphi.com/uploads/short-url/sJlv0EIpVTo3jzyHGeTyMSwrbBX.pdf · CAD & Robot Catalog release (CD) · CAD (Google Drive) · Robot Catalog (PDF) · Statbotics 2231 · The Blue Alliance 2231

**#34 (study pick)** — **5000 Hammerheads** Icarus — Marblehead, MA — **EPA 236** (41-23)

### Turreted vision-tracking fuel scorer; rack-and-pinion roller intake; dye rotor; swerve

| | |
|---|---|
| **SHOOTER**<br>Turret-mounted flywheel shooter with adjustable hood; camera-tuned auto-aim. Shot parameters (hood angle + wheel speed) come from InterpolatingTreeMap lookup tables with empirically-measured time-of-flight. | **INTAKE**<br>Rack-and-pinion deploy driven by Kraken X44 motors (on TalonFX controllers), ~48:17 & 50:17 belt reductions; spin rollers |
| **INDEXER / HOPPER**<br>Dye rotor plus a spin motor and dual feed motors (feed + follower); unjam logic with hook/feed stall detection | **DRIVETRAIN**<br>Swerve (CTRE, Kraken motors on TalonFX controllers) |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Shoot-on-the-move: 5 iterations of time-of-flight recursion with velocity- and latency-compensated turret tracking; robot auto-starts scoring ~2s before an active shift and stops ~1s after. |
| **VISION**<br>6x Arducam OV9281 on PhotonVision (front L/R/center + back L/R/center), MegaTag; welded-field AprilTag layout used at Worlds | **PUBLIC CAD / DESIGN**<br>Championship CAD (Fusion 360) + open GitHub code |

**LESSONS FOR STUDENTS**

- The Open Alliance build thread, Fusion 360 CAD and open GitHub code (with a TurretIOTalonFX subsystem) document a turret robot end-to-end. That is a manageable scope for smaller teams.

**Fusion 360 CAD:** https://a360.co/4wuUiK0 · Championship CAD (Fusion 360) · Tech binder (Icarus 2.0) · Open Alliance build thread · GitHub (2026Rebuilt)

**#36 (study pick)** **359 Hawaiian Kids** Poi Pounder XXVII (V1 & V2) — Hawaii, USA — **EPA 233** — 69-11

## Turret + adjustable-hood flywheel shooter; slapdown intake; two-stage indexer; sliding hopper; climber; swerve

| | |
|---|---|
| **SHOOTER**<br>Adjustable-hood flywheel on a turret. Flywheel: 2x Kraken X60 (1:1) spinning three 4" 60A Stealth wheels + two trimmed SDS 4" brass flywheels (added moment of inertia), with five 1" Sushi accelerator rollers and a 3D-printed back ramp to compress the FUEL. Hood: Kraken X44 (167.5:1), 3D-printed sector-and-pinion, 28 deg range. Turret: 7.5" ID X-Contact bearing, custom 8ST/10DP gear, +/-200 deg (400 deg total) on a Kraken X44 (50:1), biased to the back-left corner for shoot-on-the-move. | **INTAKE**<br>Motor-powered slapdown, three full-width horizontal rollers, touch-it-own-it. Deploy: Kraken X44 (50:1) via #25 chain on a coaxial shaft (also drives the floor belts) plus a spring-loaded kicker bar; the deployment motion also actuates the prismatic sliding hopper. Rollers: Kraken X60 (1.60:1), 1.5" polycarb tubes in silicone; 1/4" pocketed 7075 side plates, SRPP front guard. (V2 rebuild swapped this slapdown for a linear intake whose plates directly carry the extending hopper.) |
| **INDEXER / HOPPER**<br>Two stages. Tunnel (Kraken X60, 1.88:1): top + bottom conveyors, two 15 mm belts each, align the FUEL into one controlled stream. Accelerator (Kraken X44, 1.46:1): three 2" Stealth + custom-lathed 2.5" Colson wheels fling the FUEL up into the turret. Fed by a sliding/extending hopper and a floor conveyor (back kicker roller -> Kraken X60, 6.30:1). | **DRIVETRAIN**<br>SDS Mk5n R3 swerve (4 modules), Kraken X44 drive + steer, 26.09:1 steering, 4" molded spike-grip wheels (2.25" wide), 19.2 ft/s. 28x26" pocketed aluminum frame; Pigeon 2.0 IMU + CTRE CANcoders on all four modules. |
| **ENDGAME**<br>Climber (per the binder). | **AUTONOMOUS**<br>Single-driver, two-button stateless automation: auto spin-up + auto-aim to the hub or an alliance-zone corner by field position; the hood auto-lowers under the trench. PathPlanner autos with a profiled point-to-point PID controller for precise climbs. |
| **VISION**<br>Five cameras feeding full-field AprilTag localization, logged via AdvantageKit / AdvantageScope 3D replay. Robot pose, vision targets, and trajectories are all modeled in physics simulation. | **PUBLIC CAD / DESIGN**<br>CAD for both robots (Onshape) + 28-page technical binder |

**LESSONS FOR STUDENTS**

- The two-stage indexer decouples organizing balls from launching them.
- The V1 to V2 rebuild is packaging-driven iteration: compare the V1 rail-carriage hopper with the V2 directly-coupled hopper.
- Physics-based simulation sped iteration and de-risked hardware before the robot existed.

**CAD release:** https://www.chiefdelphi.com/t/team-359-hawaiian-kids-2026-cad-release-q-a/521575 · CD CAD release + Q&A · Statbotics · The Blue Alliance

#32 (study pick)

## 694 StuyPulse HAL

New York, USA

**EPA 229**

46-13

### Hooded single-flywheel turret shooter, ground intake, vertically-expanding hopper, L1 side-hang climber; WCP x2i swerve

| | |
|---|---|
| **SHOOTER**<br>Hooded single-flywheel shooter mounted on a turret. Pre-season prototyping compared double-flywheel and single-flywheel overhand/underhand shooters; they built a hooded single-flywheel plate set with a polycarb sheet curved around the plates and a slot to adjust top-roller compression. The CAD has a 'Shooter + Turret' subassembly (33 features). | **INTAKE**<br>Ground intake; the team reused their 2025 coral ground intake for early testing and planned a wider intake to enlarge the hopper footprint. CAD has 'Intake' and 'Conveyor' subassemblies. |
| **INDEXER / HOPPER**<br>Conveyor feeds fuel from the intake to the turret; the CAD has 'Indexer' (8) and 'Conveyor' (4) features. The design goal was to intake and shoot simultaneously via the turret. | **DRIVETRAIN**<br>Swerve on WCP x2i modules, raised using the module's top-mount style (tubing sandwiched above the main plate with a WCP top plate) to gain ~1in clearance over the field bump; Kraken drive motors. The alpha/Kitbot used older MK4i modules from past robots. |
| **ENDGAME**<br>L1 side-hang climber (CAD 'Climber', 8 features). The team deprioritized climbing versus hopper capacity but kept an L1 climb to preserve the endgame ranking point; an L3 climb was judged worth only a fraction of a cycle in this game. | **AUTONOMOUS**<br>The turret was chosen partly for more time-efficient autos (shoot while intaking regardless of heading). |
| **VISION**<br>Limelight (the CAD includes 'Limelight Views'; the team's plan stated adding a Limelight). | **PUBLIC CAD / DESIGN**<br>CAD (Onshape) - HAL |

**LESSONS FOR STUDENTS**

- 694 documented the turret pros/cons against the added complexity and wire management, and chose to build one. Read it as a worked example of the decision.
- The hopper lid stays below trench height (about 22in) when empty and passively rises toward the 30in height limit as fuel fills the robot, adding a layer of storage without blocking trench access.

**Onshape CAD:** https://cad.onshape.com/documents/2b44ae58b051bc8b70f03c29 · Open Alliance build thread (CD) · CAD - HAL (Onshape) · Code (GitHub, StuyPlus-2026) · Statbotics 694 · The Blue Alliance 694

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#36 (study pick)

## 1833 Team BEAN Clover

Georgia, USA (Peachtree)

**EPA 225**

60-8

### Laned shooter with a floor intake and a kicker-plate feed

| | |
|---|---|
| **SHOOTER**<br>Laned shooter (rather than a drum). Before Worlds they weighed rebuilding from laned shooters to a drum and decided to keep the lanes, citing time, budget, and a judgment that other subsystems and driver practice would yield more. | **INTAKE**<br>Floor intake, reworked before Worlds. Prior to Champs they were unable to intake from the depot, so they invested in the intake ahead of Newton. |
| **INDEXER / HOPPER**<br>Kicker-plate feed moves fuel to the shooter. They had issues with the strength of the kicker plates before Worlds. | **DRIVETRAIN**<br>Unconfirmed |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Won the Autonomous Award (Google) at both Gainesville and the Peachtree District Championship. |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>CAD (Onshape) |

**LESSONS FOR STUDENTS**

- Scoping late-season changes to expected payoff: compare the payoff of rebuilding with other work.
- Read the public code repo for autonomous structure.

**Onshape CAD:** https://cad.onshape.com/documents/11b1fc83786708b81b063e5c/w/a7854643bff51f67974cba0b/e/82ecb4d9374be4387f3eecba · CAD + code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics 1833 · The Blue Alliance 1833

## #45 5907 CC Shambots Chirp

Michigan, USA (FiM)

**EPA 224**

48-19

### Four-motor flywheel shooter, 10-40 deg adjustable hood, extending roller intake, three-stage indexer; swerve

**SHOOTER**

Four flywheel motors on CAN 31-34, configured as two counter-rotating pairs, on a 1:1 drum ratio with a 2.125 in flywheel radius and a 3000 RPM ceiling. Velocity tolerance is 6 rotations per second and modelled flywheel efficiency is 0.9. The hood is a separate motor on CAN 35 through a 1/9 x 10/160 reduction, travelling 10 to 40 deg under Motion Magic with a 2 deg tolerance. Shot parameters come from an InterpolatingTreeMap with 17 breakpoints between 1.0 m and 5.0 m of hub distance, and a second table covers passing shots. A latency coefficient of 0.5 compensates the shot for robot motion. TalonFX is the controller. The shooter side plates were pocketed by hand with no FEA and each lost more than half its starting weight; the shooter pivot uses an aluminium bushing machined in house instead of a hex bearing to cut friction.

**INTAKE**

Two roller motors on CAN 11 and 12 plus one actuator motor on CAN 13. The actuator extends the intake from 2.0 in retracted to 11.493 in, under Motion Magic with a 0.5 in position tolerance, and has two further named positions at 0.0 in and 6.0 in used for agitation. Roller speed is 85 rotations per second.

**INDEXER / HOPPER**

Three driven stages, each its own motor: a floor stage on CAN 21 at 80 rotations per second, a tunnel bottom stage on CAN 22 at 75, and a tunnel top stage on CAN 23 at 85. The intake actuator positions double as an agitation cycle, so there is no separate agitator mechanism in the released code.

**DRIVETRAIN**

Swerve generated by the CTRE Tuner X swerve project and run on Phoenix 6. Drive reduction 7.03125:1, steer reduction 26.0909:1, 2 in wheel radius, free speed 5.12 m/s at 12 V. Module centres sit at 11.125 in longitudinal and 10.625 in lateral, giving a 22.25 x 21.25 in module footprint. Steer feedback is a fused CANcoder, the IMU is a Pigeon 2, slip current is set at 120 A, drive supply limit 50 A and steer stator limit 60 A.

**ENDGAME**

The released code contains no climber subsystem. The Statbotics breakdown attributes 67.3 of the team's 223.8 EPA to endgame fuel, so the endgame is scoring rather than a mechanism.

**AUTONOMOUS**

PathPlanner, with the robot configuration loaded from the GUI settings file. The AUTO_PATHS array in the public release is empty, so the routines themselves are not in the release. The team won the Autonomous Award at the FiM Woodhaven district event.

**VISION**

PhotonVision with two forward cameras, pv_FrontLeft and pv_FrontRight, mounted 13.758 in forward, 3.375 in either side of centre and 13.145 in high, pitched 20 deg and yawed 10 deg outboard. Each camera has a 16 ft trust cutoff. Measurements are screened by an ambiguity threshold of 0.4 averaged over 100 frames with a distance scalar of 2. The AprilTag layout loaded is the welded field, so tag poses match the welded field only.

**PUBLIC CAD / DESIGN**

Onshape CAD (robot Chirp)

**LESSONS FOR STUDENTS**

- Three independently tuned indexing stages are a cheaper way to control ball spacing than one long driven path.
- Vision trust is set per camera rather than globally, and measurements are screened on a running average of ambiguity instead of a single frame.
- A team copying this code for an AndyMark field has to change the AprilTag layout. The field dimension comments in Constants list both.
- The machined pivot bushing was chosen for friction and for in-house capability rather than part count.

**Onshape CAD:** https://cad.onshape.com/documents/4bcf78cab5066972a785c135 · 5907 CC Shambots CAD and Code Release (CD) · Onshape CAD · Code (GitHub) · Statbotics 5907 · The Blue Alliance 5907

| #48 (study pick) | **604 Quixilver** Toploader<br>San Jose, California, USA | EPA 222<br>47-16 |
|---|---|---|

## Single-stream turret shooter fed by a rotary 'dye rotor' serializer; polycarbonate intake arms; sliding hopper; swerve

**SHOOTER**
Single-stream turret. The team chose a turret over a full-width launcher to score and feed on the move, accepting a balls-per-second reduction: a prototype serializer showed about 15 BPS single-stream against about 23 BPS full-width. The turret is packaged to minimise rotationally-swept volume, which preserves hopper capacity and ball flow into the bucket. The turret slider bearing is a stack of polycarbonate and aluminium plates; 3D printed PETG spacers between the ring and gear set bearing clearance by trial and error without remaking major parts, and the joint is heavily greased. Launch parameters come from an offline Monte-Carlo sweep of angle and velocity pairs, saved to a CSV table, selected for least sensitivity to error; launch vectors are then adjusted to cancel the robot's field-plane velocity for shooting on the fly.

**INTAKE**
Independently driven intake arms, one Kraken X44 per side to deploy and retract, which fixed a chain-breaking problem in the mechanically coupled 'slap down' intake on their 2024 robot; the arms can flex endstop to endstop without damage. Motors stay inside the frame perimeter and drive the rollers through belts, with 3D printed TPU 'reverse pants' on each arm holding the belt up under deflection to prevent derailing. Arms are polycarbonate rather than SRPP, chosen to avoid permanent deformation that would change belt centre-to-centre distance and lose tension. Dead-axle rollers: polycarbonate rollers covered in silicone tubing, TPU pulley plugs, an individual dead axle per plug. A single-deploy kicker bar is spring-loaded with surgical tubing, held by TPU tube plugs and string across the assembly to avoid bolted stress concentrations.

**INDEXER / HOPPER**
'Dynamic Point of Capture' (DPC) serializer: a rotating mechanism, commonly called a dye rotor and inspired by paintball gun feeds, spins to scoop fuel out of a circular bucket, bringing the capture point to the fuel rather than the reverse. Rotation and up-feed run on independent motors so the ratio between them can be tuned in software. Omni wheels on the side feed rollers reduce forces and improve efficiency. Fuel is guided to the turret through a tube formed from corrugated plastic with 3D printed ends, each ball pushing the previous one up into the turret. The plywood bucket is split into two segments for battery and electronics access; plywood was chosen for stiffness-to-weight. The sliding hopper is two laminated layers of corrugated plastic, deployed by the intake and retracted by surgical tubing, with 0.090 in polycarbonate sandwiched around the slots; panels can be cut with scissors and are joined with tape.

**DRIVETRAIN**
Swerve. 27 in square chassis, sized to maximise the area available for the circular serializer bucket without custom modules or chamfered corners. Internal cross-bars form isolated battery and swerve compartments to protect electronics. Aluminium 'armour' integrated with the rear bumper mounting hardware protects the serializer from impacts inside the frame perimeter.

**ENDGAME**
Unconfirmed

**AUTONOMOUS**
Paths are pre-planned trajectories from Quikplan, the team's time-optimal swerve trajectory planner. Paths are designed around bump-only limitations and alliance-partner compatibility, using strategic pauses to avoid collisions and exploit gaps in traffic. Key path features including pauses can be edited up to the last minute without redeploying code. Alliance-partner match video is timestamped and compared against robot simulation to check compatibility. Won the Autonomous Award in the Newton Division.

**VISION**
PhotonVision MultiTag, four cameras (one on each side, two on the back). Each camera generates a robot pose; the software picks the pose from the camera seeing the most hub AprilTags, then the most total tags, and discards poses with nonzero ambiguity or an old timestamp. The turret is only considered ready to launch when velocity, launch angle and yaw are in tolerance and the cameras have a recent image of at least two tags.

**PUBLIC CAD / DESIGN**
Onshape CAD

**LESSONS FOR STUDENTS**

- Design for the shop that exists: the stated constraints were tubing jigs, Fabworks, 3D printing and minimal router access.
- Launch parameters solved offline keep onboard computation low.
- Automation was scoped so that one driver runs the robot: single-button scoring, automatic target selection by field zone, and automatic serializer start and stop based on turret readiness.

**Onshape CAD:** https://cad.onshape.com/documents/0baaf96036e07596972c4d48/w/7236f4fc3d9cbd0caa07ab6c/e/e54d06e7bb840dd08bd9d927 · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Statbotics 604 · The Blue Alliance 604

## #55 8214 Cyber Unicorn — EPA 217

China — 16-8

### Sheet-metal turret shooter; roller intake; indexer/hopper; swerve

| | |
|---|---|
| **SHOOTER**<br>Custom CNC-machined flywheel (bore geometry designed to be cut with a 2.5mm end mill); turret-mounted shooter | **INTAKE**<br>Roller intake |
| **INDEXER / HOPPER**<br>Indexer/hopper present | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>Full CAD release (Onshape) |

**LESSONS FOR STUDENTS**

- Design parts around your tools (DFM): the flywheel bore was shaped so the end mill could cut it.
- Outsource sheet metal to a sponsor and machine the rest in-house. That division of labor builds a clean, precise robot fast.
- Tidy wire management and packaging correlate with reliability and top EPA: visual order is an engineering signal.

**CAD release:** https://www.chiefdelphi.com/t/frc-cyber-unicorn-8214-2026-champ-robot-cad-release/521310 · CD champ-robot CAD release · Statbotics

## #56 (study pick) 4481 Team Rembrandts — EPA 214

Eindhoven, NL — 37-15

### Turret hooded flywheel; passive linear intake; 'washing machine' spindexer; swerve

| | |
|---|---|
| **SHOOTER**<br>Turret hooded flywheel, their 4th shooter design: a static back wall (they tried twin rollers but found them unneeded for high BPS); Teflon tape to cut friction/backspin; a custom flywheel to hold RPM; a switch to metal gears for champs (the 3D-printed ones damaged easily). Dual NEO Vortex single axle, ~370 deg travel, herringbone rack-pinion hood 15-45deg [BtB transcript] | **INTAKE**<br>Passive linear intake (slides in/out, doesn't fold). No deploy motors: constant-force springs + gravity drop it at match start. It can't retract mid-match (deemed unnecessary). Impact-resistant: it can fold in on a front hit without breaking [BtB transcript] |
| **INDEXER / HOPPER**<br>'Washing machine' spindexer that spins fast. Iterated to a larger outer chamber + inner cone (helps balls settle -> higher BPS), with the gearbox + two motors inside the cone. Lowered the walls to admit balls easier. Turret cable chain kept taut by an elastic band (shortest at the anchor, stretches as it spins; never snags) [BtB transcript] | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Vision auto-aim for the turreted hooded flywheel. |
| **VISION**<br>Four cameras at different angles to approximate robot pose. Focus on power management: per-subsystem current limits + supply voltage [BtB transcript] | **PUBLIC CAD / DESIGN**<br>Public CAD (GrabCAD) + Open Alliance thread |

**LESSONS FOR STUDENTS**

- The gear-ratio documentation (45.7:1 flywheel, 58.5:1 hood) is a model for sizing reductions to control-vs-speed needs.
- A through-bore encoder on the hood enables closed-loop angle control, not open-loop presets.
- Turret cable-sleeve fix (lengthening it to go from ~270 deg to full travel): design the wire path for full range from the start.
- The team authored the '3DM' game-analysis document: read it.

**GrabCAD:** https://grabcad.com/team.rembrandts.4481-1 · Open Alliance thread · GrabCAD · Behind the Bumpers (full transcript)

#62 **3476 Code Orange** Sandspit
Irvine, CA

**EPA 212**
54-9

## Full-width drum shooter (no turret), ~15 BPS; four-bar intake; pleated-belt roller floor [BtB transcript]

| | |
|---|---|
| **SHOOTER**<br>Full-width drum shooter (~15 BPS). Two ~5in in-house-machined flywheels: steel compound, mass at the outer radius for max inertia-per-weight, cut on a Haas mini-mill over a multi-day, multi-tool process. Active hood on an in-house 10DP gear, symmetrical with a shaft + a 1:1 through-bore encoder for shoot-and-move. 3D-printed hood backings [BtB transcript] | **INTAKE**<br>Four-bar intake with active ball control: its linear push turns the intake into a ramp funneling fuel toward the shooter. A 'scotch yoke' makes the hopper box follow the intake up/down. Custom-machined 6061 sprockets (stock shortage); pivot driven through a 1in round tube to both ends. Tried coaxial roller power (balls caught) -> switched to Citrus's double-motor intake (much better) [BtB transcript] |
| **INDEXER / HOPPER**<br>A 2:1 gearbox drives the roller floor + the first feeder roller. 'Pleated belts' (TPU-printed cleats) on 3D-printed CF crown rollers agitate + push balls linearly down the ramp (replacing two top rollers that let balls fall back). Crown belts drive the whole floor without belts between every roller (less friction). An 'epoxy coupler' roller mount (epoxy + embedded 4-40 nuts) cuts ~50% of stub-roller weight + allows quick roller swaps [BtB transcript] | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>PhotonVision (switched from Limelight) with six cameras: four AprilTag (two forward, two back) + two rear object-detection cameras that find the densest fuel. 1600x800 with a custom ML algorithm that crops each frame to the AprilTags so it computes only on them (keeps ~50 fps and a stable pose). Two Rubik Pi 3 co-processors [BtB transcript] | **PUBLIC CAD / DESIGN**<br>Reveal video only (no downloadable CAD) |

**LESSONS FOR STUDENTS**

- Anything a ball can sit on, wedge above, or spin on is the enemy of ball flow.
- Dividers/lanes that look organized often trap balls. Smooth continuous paths beat segmented ones.
- Make the feeder transparent and film with high-speed cameras to diagnose flow problems you can't see at speed.
- Use the intake/box as a compactor to push fuel into the feeder while shooting.

**CD thread:** https://www.chiefdelphi.com/t/team-3476-code-orange-2026-sandspit-robot-reveal/516875 · CD reveal + flow philosophy · Reveal video · Behind the Bumpers

#71 (study pick)

## 195 The CyberKnights Big Daddy V2 (Biggest Daddy)

Southington, CT

EPA 206

45-21-1

### Drum shooter (swapped from a 4-lane shooter mid-season); over-bumper slapdown intake; moving floor; uptake; swerve

| | |
|---|---|
| **SHOOTER**<br>4in titanium drum + a 1:1 brass flywheel + two carbon-fiber back rollers at 2:1; ~25 BPS. Started as a 4-lane shooter (~15 BPS), but balls jammed on the lanes -> removed the lanes and went to the drum [BtB transcript] | **INTAKE**<br>Over-bumper slapdown deployed off an X60 (planetary 9:1 + two 3:1 for a fast deploy, esp. in auto). 3 polycarb rollers: a bottom kicker (no silicone; it caught on the bump) + two silicone front rollers; one Kraken X60 ~3:1. Swapped from an earlier rack-and-pinion + 4-roller intake that stuck/broke on hits (the new one is lighter) [BtB transcript] |
| **INDEXER / HOPPER**<br>Moving floor = WCP stub-roller system (carbon-fiber rollers with silicone) on one Kraken X60 (~1:1, pushes balls hard into the uptake); uptake = six 1in carbon-fiber silicone tubes on 2x Kraken X60 [BtB transcript] | **DRIVETRAIN**<br>Swerve (auto-shoot puts the swerve in X-mode to resist defense) [BtB transcript] |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Four Arducams -> a Mac Mini running PhotonVision. If any camera sees 2 AprilTags, it fuses with odometry for full field awareness. Runs ROS 2 (a node per subsystem, so one crash doesn't take down the robot) + Molex locking connectors. Full match logging (all node messages + 4 camera feeds) for replay [BtB transcript] | **PUBLIC CAD / DESIGN**<br>Full CAD + code + docs (Onshape + SolidWorks) |

**LESSONS FOR STUDENTS**

- Diagnose and fix recurring failures. They cured belt-snap by swapping to chain at the highest-load point.
- The custom CyberKraken power/CAN boards use Anderson connectors, chosen over Molex SL for shock resistance. The electrical design is documented in detail.
- Win by refining a proven architecture through continuous in-season optimization.
- Study the public ROS-based robot code as an advanced software target.

**CAD release:** https://www.chiefdelphi.com/t/team-195-the-cyberknights-2026-cad-code-and-documentation-release/521390 · CD CAD/code/docs release · Onshape CAD · Behind the Bumpers · GitHub

#79

## 6328 Mechanical Advantage Darwin

Littleton, MA

EPA 204

51-13

### Turret shooter; AdvantageKit + Northstar vision; swerve

| | |
|---|---|
| **SHOOTER**<br>Turret shooter with adjustable 3D-printed (PLA+) hood; 4in WCP Fairlane main wheel + 2in solid-core flex top wheel on a live single shaft; optional brass flywheel for added inertia; hood gear routed from WCP SRPP | **INTAKE**<br>Compliant-wheel roller intake (prototyped publicly) feeding a centering/serializing path |
| **INDEXER / HOPPER**<br>Intake -> centering hopper -> tower uptake -> turret/flywheel; serialization handled in software so balls feed one at a time | **DRIVETRAIN**<br>Swerve; .090/.063 5052 aluminum sheet with lightening tricks |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>AdvantageKit + Northstar AprilTag localization (Orange Pi); auto-aim and autonomous routines in the released code. |
| **VISION**<br>Northstar custom AprilTag pipeline on Orange Pi coprocessors. Alpha camera mounted behind the shooter hood for turret-to-hub alignment. Full AdvantageKit/AdvantageScope logging + sim. | **PUBLIC CAD / DESIGN**<br>Open + code (AdvantageKit); CAD links in build thread |

**LESSONS FOR STUDENTS**

- Study the AdvantageKit IO-layer architecture and its deterministic input->output ordering.
- They openly teach routing custom gear teeth for the turret and hood.

**Code (GitHub):** https://github.com/Mechanical-Advantage/RobotCode2026Public · CD build thread · Reveal 'Darwin' · GitHub (public code)

#80 **3255 SuperNURDs** The Thing
California, USA
**EPA 202**
37-23

## 1678/254-style 'big dumper' (rebuilt to it in week 2): steel-drum flywheel, aluminum kicker/hood rollers; swerve

| | |
|---|---|
| **SHOOTER**<br>Big-dumper flywheel scorer. Deliberately does not use an inertia (mass) wheel; a 4 in steel drum serves as the flywheel, storing energy in the drum mass itself, with 2 in aluminum drums acting as the kicker/hood rollers that meter and launch FUEL. Converted to this layout mid-season (week 2). | **INTAKE**<br>Unconfirmed |
| **INDEXER / HOPPER**<br>Unconfirmed | **DRIVETRAIN**<br>Swerve. 'Datum' anti-defense hardstops let the robot quickly align to the hub, brace against the trench, and resist being pushed. |
| **ENDGAME** | **AUTONOMOUS** |
| **VISION**<br>3x Limelight 4. | **PUBLIC CAD / DESIGN**<br>CAD release (GrabCAD) |

**LESSONS FOR STUDENTS**

**GrabCAD:** https://grabcad.com/library/frc-2026-team-3255-the-supernurds-1 · CD - CAD Release (Team 3255) · GrabCAD - The SuperNURDs · Statbotics 3255 · The Blue Alliance 3255

#87 (study pick) **1156 Under Control** Tempest
Brazil
**EPA 198**
48-12

## High-capacity turret shooter: adjustable-hood flywheel on a +/-250 deg turret; spindexer; feeder; climber; swerve

| | |
|---|---|
| **SHOOTER**<br>Turret-mounted adjustable-hood flywheel. Hood 15-55 deg via rack (one NEO 2.0, 15.37:1 MAX Planetary). Flywheel: two Kraken X60 (1.4:1) driving two brass inertia flywheels (~150 g ea) + two 4 in urethane (WCP 60A). Turret: 6.5 in X-contact bearing, 42.24:1 (96T/10T), one Kraken X44, +/-250 deg.<br>Shoot-on-the-move via calibrated interpolation maps (distance -> hood angle / RPM / flight time), 20 iterations per cycle. | **INTAKE**<br>'Touch-it-own-it' ground intake. 8 mm 7075 towers + 1/2 in polycarb arms (10DP rack); front 2.25 in (AM 35A) + rear 1.25 in rollers driven by two NEO Vortex (2:1, ~33 ft/s). Rack expansion by one NEO Vortex (5:1) deploys in under 1 s; 45 deg polycarb ramp. |
| **INDEXER / HOPPER**<br>Hopper -> spindexer -> feeder. Hopper: polycarbonate + carbon-fiber; expands ~12 in to hold up to ~55 FUELS. Spindexer: 3D-printed cone, two NEO Vortex (4:1), keeps pieces moving (anti-jam). Feeder: one Kraken X60 (2.2:1), 16 compliant rollers, multi-piece flow with no dead zones. | **DRIVETRAIN**<br>Custom SDS MK4n swerve on a 28 x 26 in frame (2 x 1 in Al tube, 2 mm bellypan). L1+ (14.2 ft/s); Kraken X60 drive + Kraken X44 steer (field-oriented); machined hubs with TPU treads for grip. |
| **ENDGAME**<br>Climber (binder-documented). | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>4-camera fusion: 2 Limelight + 2 Arducam -> MultiCameraPoseEstimation with AprilTag corrections feeding odometry. Turret angle solved by trig + Chinese Remainder Theorem on two absolute encoders for a 720 deg usable range. | **PUBLIC CAD / DESIGN**<br>Full CAD (Onshape) + 32-page technical binder |

**LESSONS FOR STUDENTS**

- The 3DM process uses a priority list and the archetype decision matrix. Example functional requirements: under 22.25 in to clear the trench, and under 21 kg modules for air travel.
- The Chinese Remainder Theorem method yields the turret range without re-zeroing. Study it as a controls trick.
- The shoot-on-the-move maps are experimentally calibrated, not real-time physics. This is a practical approach to moving shots.

**Onshape CAD:** https://cad.onshape.com/documents/96bff94aa06158572d9954cd/w/156a28a4c439f1f7e5f3bf04/e/3a4d945dab8b9e666f1c061c · CD CAD + tech binder release · 32-page technical binder (Drive) · Statbotics

#93 (study pick)

## 5987 Galaxia (i.m.o. David Zohar)

Petah Tikva, Israel

**EPA 197**

6-6

### Turret-on-hood shooter fed by a spindexer; physics-model firing solutions; swerve

| | |
|---|---|
| **SHOOTER**<br>Turret-mounted flywheel + adjustable hood. Firing solution (wheel speed, turret/hood angle) from a physics model (air resistance, Magnus force) instead of a lookup table. Auto-opens hood after trench from pose/velocity. Kraken X44; SOTM in progress. | **INTAKE**<br>Floor intake on revised roller geometry tuned for rapid multi-ball pickup without jamming; used for center-field steals in auto |
| **INDEXER / HOPPER**<br>Spindexer -> curved ramp -> gate/kicker wheels. HDPE hopper walls (replaced polycarb) to reduce sticking. Counter-rotating mecanum agitator (paintball-hopper inspired) forces balls down. 2x CANrange sensors; ~55-ball goal. | **DRIVETRAIN**<br>WCP SwerveX2c; Pigeon 2.0 IMU; PathPlanner autos for neutral-zone steals |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Physics-model firing solution; hood auto-opens after the trench from pose/velocity. |
| **VISION**<br>PhotonVision on Orange Pi with AprilTags around the central target; local 3D pose algorithm | **PUBLIC CAD / DESIGN**<br>Full Onshape CAD (Open Alliance) |

**LESSONS FOR STUDENTS**

- Re-tuning the physics model means changing a few parameters, not re-populating a table.
- Borrow cross-domain ideas. The agitator is one example.
- Program in simulation first so new students can write/tune code before the robot exists.

**CD thread:** https://www.chiefdelphi.com/t/frc-5987-galaxia-2026-build-thread-open-alliance/505340 · CD build thread · In-season subsystem post · GitHub

#96

## 5406 Celt-X CX26

Hamilton, ON

**EPA 196**

45-15

### Turret single-file shooter; wide touch-it floor intake; flow-through hopper; no climber; swerve

| | |
|---|---|
| **SHOOTER**<br>Turret-mounted flywheel, single-file feed. ~8 balls/sec peak for ~2s, then ~5 sustained. Re-geared flywheel pulleys (15:30 -> 22:24) for higher speed. Used motion-profile / current control (6328-inspired) to tame large startup PID error. | **INTAKE**<br>Wide touch-it-own-it floor intake (3 balls wide); ran ~50% speed to avoid ejecting balls over the hopper; also acts as a compactor |
| **INDEXER / HOPPER**<br>Flow-through hopper (~45+ balls) with spindexer-style feed; throughput drops as far-from-spindexer balls settle slowly; adding agitators | **DRIVETRAIN**<br>Swerve; emphasis on under-trench driveability + fast cycles vs double-wide shooters |
| **ENDGAME**<br>No climber: their strategy analysis flagged climbing as low-value. | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>AprilTag targeting expected. | **PUBLIC CAD / DESIGN**<br>Onshape CX26 assembly (Open Alliance) |

**LESSONS FOR STUDENTS**

- Build an Alpha bot to surface integration problems early (melting sidewalls, chain backlash, hopper dead zones) before the comp design.
- Know your shooter's measured throughput curve. Peak BPS means little if it collapses once near-spindexer balls are gone.
- Flywheel PID from a dead stop oscillates, so plan the startup control.
- Single-wide shooters lose a measured throughput race to double-wide shooters. Compensate with the turret, driveability and faster cycles.

**CD thread:** https://www.chiefdelphi.com/t/celt-x-5406-2026-build-thread-open-alliance/511200 · CD build thread · Alpha-bot throughput analysis · GitHub

#100 (study pick)

## 971 Spartan Robotics Mixtape

Mountain View, CA

**EPA 196**

36-28

### Double-turret flywheel shooter w/ lead-screw adjustable hoods; ground intake; spindexer; swerve

| | |
|---|---|
| **SHOOTER**<br>Double turret: two flywheel shooters, each with an adjustable hood driven by a lead screw. The lead screw was chosen to minimize the turret's swept radius: it tucks into the empty 'sides' instead of making the shooter longer. One iteration let the screw creep upward and lock the hood down; fixed by tapping and bolting the bottom of the screw (fix not shown in CAD). | **INTAKE**<br>Ground intake feeding the hopper/spindexer. |
| **INDEXER / HOPPER**<br>Simple spindexer feeding both shooters. Team reasoning: at high BPS it is easier to index into two shooters than one. They considered reviving their 2017 dye rotor (similar ball-shooting game) but judged it too finicky after losing the mentor talent behind the 2017 design. | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Auto routines in the released code (Java, a break from the historic C++ stack). |
| **VISION**<br>Custom in-house vision pipeline (971 tradition). | **PUBLIC CAD / DESIGN**<br>Full robot CAD (Onshape) + code (GitHub, frc971/971-second-robot-2026) |

**LESSONS FOR STUDENTS**

- Institutional knowledge is a design constraint. Study the dye rotor case.
- This is the first year of public CAD from the team.

**Onshape CAD:** https://frc971.onshape.com/documents/cabaa0c1c77517916df80783/w/48cb057db38a03cc202ff44e/e/96844befd4591dac162d657b · CAD + Code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics

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#103 (study pick)

## 341 Miss Daisy Miss Daisy XXIV

Ambler, PA

**EPA 194**

52-9

### Hopperless 360+ turret shooter with a 4-ball-wide intake and a direct 4-2-1 ball path

| | |
|---|---|
| **SHOOTER**<br>Flywheel + fully adjustable hood (team's first) on a 360+ degree turret; fed straight from intake to shooter (no storage hopper). The turret is driven by a single 10DP gear at the centre of the turret gearing plate. Two 3D printed gears on the outer sides of that plate do not drive it: their tooth counts are coprime and each carries an absolute encoder, so turret rotation is read as a vernier sliding scale and multiple turns can be tracked. | **INTAKE**<br>4-ball-wide bottom roller intake feeding straight into the 4-2-1 path; there is no storage hopper to buffer it, so intake width is what sets the cycle rate. |
| **INDEXER / HOPPER**<br>No hopper: a '4-2-1' path narrows 4 lanes to 2 to 1 directly into the turret; ~8-ball working capacity | **DRIVETRAIN**<br>WCP X2i swerve, 27in square base, 115 lb |
| **ENDGAME**<br>Lightweight L1 climber kept deliberately cheap so mass/time goes to scoring. | **AUTONOMOUS**<br>360+ turret + adjustable hood for auto-aim; the hopperless 4-2-1 feed supports a fast continuous cycle. |
| **VISION**<br>Single turret-mounted Limelight ('limelight-turret'), MegaTag2, IMU-fused; PhotonVision IO also present; ~8.25in turret-relative camera offset | **PUBLIC CAD / DESIGN**<br>Full CAD + code + binder + pre-match checklists |

**LESSONS FOR STUDENTS**

- Going hopperless removes a storage subsystem and its jam points. It is viable if the intake-to-shoot rate is high.
- An adjustable hood lets one flywheel cover many distances.
- Reliability is a process: they publish per-match pit checklists and run a role-specialized crew.
- Multi-turn turret position is solved without a slip ring or a hard stop. It is a cheap, reproducible trick.
- Season result: 2x district event winner (Hatboro-Horsham and Centennial), undefeated at Centennial which was a team first, 154 district points before DCMP.

**CAD release:** https://www.chiefdelphi.com/t/team-341-miss-daisy-2026-cad-code-and-documentation-release/521088 · CD CAD/code/binder release · Reveal (Miss Daisy XXIV) · Robot code (public)

#123 (study pick) **9072 TigerBots** Asura + Sandstorm — Chesapeake, MD — **EPA 186** — 52-18

## Dual 720-deg twin-turret shooter (Asura); 254/1678-style 4-wide dumper (Sandstorm); swerve

| | |
|---|---|
| **SHOOTER**<br>Asura: two independent 720-degree turrets, each its own flywheel (~10 BPS combined). Sandstorm: 4-ball-wide top-level dumper/shooter (~19 BPS). Motor packaging iterated across test bots | **INTAKE**<br>Full-width roller intake feeding the indexer; intake-deploy gearbox added last |
| **INDEXER / HOPPER**<br>Asura: ~45-ball hopper. Sandstorm: ~75-ball hopper. Concluded early they'd be 'indexer-limited' and hunted the fastest indexer concepts | **DRIVETRAIN**<br>CTRE swerve, Kraken X60 drive+steer, CANcoder, 7.67:1, 1.95in wheels, ~4.0 m/s |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>3x Limelight (left/center/right) via MegaTag | **PUBLIC CAD / DESIGN**<br>CAD + code for BOTH robots (Onshape/GitHub) |

**LESSONS FOR STUDENTS**

- Prototype the game piece fast to find your binding constraint. They proved intake/shoot were easy.
- Mine prior art: they studied 254 & 1678 plus RI3D/Open Alliance before committing.
- The dual-turret robot is buildable by a young team if you stage it: test bot first, then turret controls + tensioning.

**CAD release:** https://www.chiefdelphi.com/t/team-9072-tigerbots-2026-cad-and-code-release/521421 · CD CAD + code release (both bots) · Team robot page · Robot code (public)

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#125 (study pick) **2767 Stryke Force** — Kalamazoo, MI — **EPA 186** — 55-12

## Single shoot-on-the-move turret; roller intake; spindexer; DIY swerve

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|---|---|
| **SHOOTER**<br>Single turret with a 'breadbox' adjustable hood; 3D-printed turret bearing; inverted motor mount; built for shoot-on-the-move | **INTAKE**<br>Roller intake (hardened States->Worlds; no major redesign) |
| **INDEXER / HOPPER**<br>Spindexer feeding the turret; intentionally flexes out of the way for battery changes (ergonomics); no drum | **DRIVETRAIN**<br>DIY in-house 'Third Coast' swerve on a Chrome-Moly frame; many parts 3D printed |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Built for shoot-on-the-move; fast-iterated vision. |
| **VISION**<br>Four cameras (two on each corner) on two Orange Pis (tucked under the batteries) running their custom 'Walleye' AprilTag software (in use since AprilTags launched); used heavily to correct the gyro/odometry in this odometry-driven game [BtB transcript] | **PUBLIC CAD / DESIGN**<br>Onshape (robot) |

**LESSONS FOR STUDENTS**

- DIY swerve design: their full mechanical design doc (Third Coast) is public.
- Iterate for robustness without scope creep.

**Onshape CAD:** https://cad.onshape.com/documents/a4cd67d32bba9211459890b2 · CD CAD · Onshape CAD · Third Coast swerve design doc · GitHub · Behind the Bumpers (full transcript)

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#127 **360 The Revolution** Rainmaker XXVI — Tacoma, WA — **EPA 186** (40-20)

## Full-width turret shooter, bang-bang flywheel, shoot-on-move; swerve

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|---|---|
| **SHOOTER**<br>Full-width turret; 6328-style bang-bang flywheel wrapped in a state machine; shot interpolation from data. SOTM capped ~0.5 m/s (rising); time-of-flight tracking. Passing/shooting auto-selected by robot position + match time. | **INTAKE**<br>Independent intake state machine (full-width feed) |
| **INDEXER / HOPPER**<br>Feeds the turret; managed within the superstructure state machine | **DRIVETRAIN**<br>Swerve; PathPlanner autos (kept partly command-based to avoid state-machine conflicts) |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Shoot-on-the-move and auto-selected passing/shooting within a 254/2910-style state machine. |
| **VISION**<br>Limelight + AprilTags (AprilTag field layout; a lazy-init bug once cost a loop overrun + ~5 hrs at field) | **PUBLIC CAD / DESIGN**<br>Onshape CAD + public GitHub |

**LESSONS FOR STUDENTS**

- A superstructure state machine is more flexible and testable than scattered commands.
- Adopt proven patterns as a whole, such as the AdvantageKit IO layer and ControlState for isolating subsystems to test.
- The CI pipeline (build/format/sim gates) and the AI first-pass review caught ~35 bugs. One was a missing switch break that would have broken aiming.
- The bottleneck is PR review / learning / robot test time, not code generation.

**Onshape CAD:** https://cad.onshape.com/documents/91500087e03b9e406a527d6d · CD build thread · Onshape CAD · GitHub RainMaker26

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#158 (study pick) **190 Gompei and the HERD (WPI)** Doomspiral (V1) + TURNOVER (V2) — Worcester, MA — **EPA 176** (78-26)

## Turret shooter with a belt kicker (V2 TURNOVER); roller intake; swerve

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|---|---|
| **SHOOTER**<br>V2 TURNOVER is a turret shooter with a belt 'kicker' that feeds/compresses the ball into the flywheel (they stayed turret while many moved to drums) | **INTAKE**<br>Kraken X60 FOC roller intake (integrated 'turnover' architecture) |
| **INDEXER / HOPPER**<br>Kraken X60 FOC rollers route fuel through the 'turnover' path into the turret | **DRIVETRAIN**<br>CTRE swerve, Kraken X60 FOC |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>3x Limelight 4 (shooter-mounted, climber, intake positions), MegaTag2 via their GompeiLib | **PUBLIC CAD / DESIGN**<br>Onshape - both robots (Doomspiral V1 + TURNOVER V2) |

**LESSONS FOR STUDENTS**

**Onshape CAD:** https://frc190.onshape.com/documents/ba8a365713d40fc24cf0b54f · CD 190 CAD release (both robots) · TURNOVER showcase · Onshape (TURNOVER) · Robot code (2k26)

| #153 (study pick) | 7415 Jaguar Robotics<br>California, USA | EPA 176<br>49-17 |
|---|---|---|

## Quad-lane shooter, linear intake, swappable hopper ('Noah's Ark'): tall ~100-fuel or short trench config; swerve

| | |
|---|---|
| **SHOOTER**<br>Fixed (non-turret) quad launcher: four lanes, each on its own Kraken X60, spin-up under 0.25 s; kept under 22 in tall to pass under the trench. | **INTAKE**<br>'Lintake': a 5+ fuel-wide linear intake with a powered kicker roller. |
| **INDEXER / HOPPER**<br>Swappable hopper (~100-fuel tall config or a short trench config, swapped in under 5 min via 8 bolts) feeding six powered floor rollers with zero dead zones. | **DRIVETRAIN**<br>23.5 x 31.5 in base on SDS MK5n swerve (geared to R2), 7075 punch tube. |
| **ENDGAME**<br>Designed around scoring rather than climbing the Tower. | **AUTONOMOUS**<br>Full-field localization + shooter automation for scoring autos; PathPlanner-based routines (binder details the autonomous + shot-automation sections). |
| **VISION**<br>4 Limelight 4 cameras for full-field localization | **PUBLIC CAD / DESIGN**<br>Full CAD (Onshape) + code (GitHub) + web technical binder |

**LESSONS FOR STUDENTS**

- The swappable hopper gives capacity when you can and a low profile to beat defense.
- An 'Alphabot' pre-season testbed surfaced problems early (inaccurate drum shot, illegal start config) before the competition build.

**Onshape CAD:** https://cad.onshape.com/documents/d7e50bc5f2f3accc3e502b3e/w/6c7c691839b17298be38e609/e/a406643d94e0f4b9284c165c · CAD/Code/Binder release (CD) · Technical Binder · CAD (Onshape) · Code (GitHub) · Statbotics

#213 **4915 Spartronics** ARTEMIS
Washington, USA (PNW)

**EPA 159**
45-23

## Turreted shooter and adjustable hood on a pivoting superstructure, intake, feeder and indexer pipeline, climber; swerve

**SHOOTER**
Two motors, lead on CAN 22 and follower on CAN 23, with a sensor-to-mechanism ratio of 0.9375 and an 80 A supply limit that is disabled by default in the configuration. The auto-aim layer caps shooter speed at 120 rotations per second and treats the shooter as ready within 4 rotations per second of target. A modelled 10.695 percent loss is applied on shooter-to-ball transfer. The hood is a separate motor on CAN 21 through an 85.33:1 reduction with a 40 A limit and a 4 deg aim tolerance.

**INTAKE**
Lead motor on CAN 15 through a 4:1 reduction with a 40 rotations per second ceiling. The autonomous constants give the intake a 213.05 mm extension beyond an 818.5 x 875.65 mm robot with 72.7 mm bumpers, which is the envelope the path planner respects.

**INDEXER / HOPPER**
A two-element pipeline. The indexer is a single motor on CAN 17 through a 5:1 reduction, capped at 22 rotations per second with a 60 A limit. The feeder that follows it is a motor on CAN 18 through a 34:9 reduction, capped at 100 rotations per second with a 60 A limit, with a ball-detection sensor and a 0.5 s debounce before the superstructure treats the robot as empty. Pipeline state changes are rate-limited to 0.2 s.

**DRIVETRAIN**
YAGSL-configured swerve. Each module uses TalonFX controllers for drive and steer with a CANcoder azimuth encoder, and the IMU is a Pigeon 2, all on one named CAN bus called Hydra. Drive reduction 6.03:1 on a 4 in wheel, steer reduction 26.09:1, drive current limit 60 A and steer 40 A, with a wheel grip coefficient of 2.255. Module centres are 12.164 in forward and 9.586 in lateral. Odometry runs at 120 Hz. A slip detector watches for 2.2 rotations per second of wheel slip over 5 debounce cycles and swaps in degraded pose standard deviations for 0.3 s. Driver speed is governed to 15 percent of maximum and 20 percent angular while shooting at the hub, and 50 percent while ferrying.

**ENDGAME**
Climber on CAN 24 through a 20:1 reduction with a 60 A limit, travelling -3.5 to 3.5 in the code's units, with velocity and acceleration limits of 10. The autonomous layer carries its own climb path constraints.

**AUTONOMOUS**
Not a fixed path library. The routines are built from named field points (tower, hub, outpost, depot, centre) with zone transitions, a pre-alignment step and a drive-to-POI command, run through a PPHolonomicDriveController with separate path constraints for drive-to-centre, bump, intake, climb and align moves. Tolerances are 1.5 cm, 3 deg and 2 in/s. The chooser composes routines rather than selecting pre-drawn paths.

**VISION**
Three PhotonVision cameras, named evan on the front tower pitched 30 deg down, val on the back tower pitched 30 deg down and facing rearward, and daniil mounted at the RIO pitched 26 deg down and yawed 297 deg. A Limelight turret camera called limelight-argos, with its own turret-to-camera transform, is present in the code but commented out of the primary camera list. Results pass a filter stage (90 ms maximum latency, 7 m maximum single-tag distance, 0.18 maximum ambiguity, area band 0.04 to 10) and then a fusion engine that requires at least two cameras inside a 20 ms timestamp window and rejects outliers beyond two sigma. Standard deviations are computed per frame from a 0.41 m base weighted by ambiguity 0.7, tag area 0.6 and latency 0.2, rather than fixed. A 0.5 s turret history buffer lets a turret-mounted camera be referenced back to the chassis. The AprilTag layout loaded is the welded field.

**PUBLIC CAD / DESIGN**
Onshape CAD (robot ARTEMIS)

**LESSONS FOR STUDENTS**

- With one processor interface, a camera can be commented out without touching the consumer.
- Slip detection is wired into the pose estimator rather than disabling odometry.
- Speed governing while shooting is a software answer to a problem often solved in hardware.
- The turret is geared about 92:1, with the absolute encoder on the final 11:84 stage. The superstructure enforces a turret window while the pivot sweeps. This is a worked example of interlocking two articulated axes.

**Onshape CAD:** https://cad.onshape.com/documents/16efabe4144263150bdc02ed · 4915 CAD and Code Release (CD) · Onshape CAD · Code (GitHub) · Statbotics 4915 · The Blue Alliance 4915

#224 (study pick)

## 3847 Spectrum Infrared

Texas, USA

EPA 158

39-27

### Turreted hood launcher, extending roller-bed intake, ball-tower indexer; swerve

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|---|---|
| **SHOOTER**<br>Turret-mounted launcher with hood. Per the blog motor budget: turret on a Kraken X44, 2x Kraken X60 launcher flywheel, hood on an X44 or servos. | **INTAKE**<br>12 in extending roller-bed intake (XM sketches): rack-gear extension (linear or slightly curved), single 2 in roller with a fold-down kicker bar, deploy motors stationary near the front frame rail. Final geometry sits 5 deg over the swerve module: the smallest angle that still compresses FUEL into the carpet (per mentor Q&A). |
| **INDEXER / HOPPER**<br>Roller bed feeds two-ball inputs through 2-to-1 gates into a single ball tower that feeds the turret; lower powered rollers plus top rollers vector the balls. Hopper walls are laser-cut polypropylene (a laserable, Loctite-safe polycarb substitute). They also published a 4-lane 'sideways spindexer' concept (4x throughput idea) they considered for the rebuild. | **DRIVETRAIN**<br>Swerve. AM (alpha) ran an existing MK4i base, with MK5n considered for space claims; later machines kept the same footprint. |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Dev-line approach (AM/XM/PM/FM): final vision/auto decisions deliberately deferred until after seeing the game at events. |
| **VISION**<br>Limelight mounted on the turret; blog discusses aiming via odometry vs turret-camera tx/ty, with AprilTags across the field providing pose updates. | **PUBLIC CAD / DESIGN**<br>CAD for all 5 robots (Spectrum CAD index page) |

**LESSONS FOR STUDENTS**

- In the dev-line approach, architecture is a season-long iteration, not a kickoff bet.
- Small geometric insights matter, as the intake angle shows.
- Study the polypropylene hopper walls as process constraints driving material choice.

**CAD release:** http://2026cad.spectrum3847.org/ · OA Build Blog (CD, 416 posts) · CAD - all 5 robots · Statbotics

#232

## 1756 Argos Miami

Illinois, USA

EPA 155

32-17

### Top-rear roller launcher; Madtown (1323)-style roller intake v2; short spindexer; MK5n swerve

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|---|---|
| **SHOOTER**<br>Top-rear roller launcher assembly, visible in the Fusion model. | **INTAKE**<br>'Madtown_Intake_v2' (92-part subassembly): a roller ground intake in the style popularized by 1323 Madtown Madness, iterated to v146 in their Fusion history. |
| **INDEXER / HOPPER**<br>'Spindex_Short': a compact spindexer, the largest subassembly in the model (397 parts). Architecture: intake -> short spindexer -> launcher. | **DRIVETRAIN**<br>SDS MK5n swerve base (93 parts) on a sheet-metal chassis with riveted side panels. |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Limelight on a dedicated rear mount ('back_LimeLight_Mount'); two camera-FOV volumes ('FOV Off') are modeled in the assembly to plan vision coverage. | **PUBLIC CAD / DESIGN**<br>Full robot CAD (Fusion 360 / A360) |

**LESSONS FOR STUDENTS**

- Modeled FOV volumes are a cheap check of vision coverage before building anything.
- The Madtown intake history shows borrow-then-refine engineering.
- The short spindexer keeps the proven anti-jam architecture while reclaiming vertical space. Compare it with full-height spindexers, such as 1756's previous seasons or 3847's concept studies.
- A complete, well-organized Fusion 360 release is rare; most teams publish Onshape.

**Fusion 360 CAD:** https://a360.co/4c8kxxw · Full CAD (Fusion/A360) · Spectrum CAD Collection · Statbotics

#252 **1714 MORE Robotics** Steel
Wisconsin, USA
**EPA 150**
43-20

## 17 in-wide rear drum shooter with adjustable hood, 4x Kraken X44; roller-floor intake; deployable polycarb blocker

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|---|---|
| **SHOOTER**<br>17 in-wide rear-mounted drum shooter with an adjustable hood, powered by 4x Kraken X44. The team chose X44 over X60 for power density (W/lb) because making weight drove the rebuild: they budgeted ~200 A for a shot (~120 A to the shooter, ~80 A to rollers/indexer), and X44/X60 give similar peak power at a 40 A limit while four X44s save ~2 lb. The earlier robot 'Flint' used a single-wide front fixed shooter (2x Kraken X60, later 2x X44) with an adjustable hood. | **INTAKE**<br>Roller-floor intake; the intake pivot was one of only two subsystems carried over from Flint. Many rollers/pulleys are 3D printed (eSun PLA+, some PETG-CF/PA-CF). |
| **INDEXER / HOPPER**<br>Hopper/indexer feeding the drum shooter; the front of the hopper is attached to the intake so they fold down together, and it was cut down to fit in frame and respect extension limits. | **DRIVETRAIN**<br>Carried over from Flint; type unconfirmed. |
| **ENDGAME**<br>A deployable polycarbonate blocker (defensive) doubles as the folded-down front of the hopper when stowed; it was redone between Worlds and MROC after shearing the poly in Curie Finals 2, widened, and held down by surgical tubing looped to the bash bar when deployed. | **AUTONOMOUS**<br>Won an Autonomous Award at the Mukwonago district event. |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>CAD (Onshape, MROC config) |

**LESSONS FOR STUDENTS**

- Do current budgeting up front to avoid brownouts.
- Study the blocker as a case of integrating defense into structure.
- For 3D-printed rollers/pulleys: solid walls, large layer height, hex-bore inserts, printed upright. Exposed parts are protected and swapped frequently.

**Onshape CAD:** https://cad.onshape.com/documents/db03fec505ded9ba473e7325 · Post-season thread + CAD (CD) · CAD (Onshape, MROC) · Statbotics 1714 · The Blue Alliance 1714

#259 (study pick) **5847 Ironclad**
Bradley, IL
**EPA 149**
31-13

## Turret adjustable-hood flywheel; roller intake; spindexer + hopper; swerve

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|---|---|
| **SHOOTER**<br>Hooded shooter with an adjustable hood. The hood was a key iteration point: broke two before settling on a wide-bottomed, more rigid design. | **INTAKE**<br>Roller intake (prototyped alongside shooter/spindexer/hopper using KrayonCAD concepting) |
| **INDEXER / HOPPER**<br>Spindexer feeding the shooter, plus a hopper; tuned in code for spindexer rotation/feeding | **DRIVETRAIN**<br>Swerve, Kraken X60 drive + Kraken X44 steer, top-mount 'brainpan' electronics; built to shoot-on-the-move |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Limelight for turret aiming + pose; code adds voltage/supply/torque inputs and tunes intake, feed, shoot, spindexer, hood angle, and turret position | **PUBLIC CAD / DESIGN**<br>Onshape (robot) |

**LESSONS FOR STUDENTS**

- Design prototypes for adjustability/versatility from one set of core parts (sliding-hood compression plate).
- The Limelight turret aiming is an accessible vision approach for a mid-tier team.

**Onshape CAD:** https://cad.onshape.com/documents/b3b80a0c9175b5a0ee25425b · CD build thread · Onshape CAD · Behind the Bumpers

#269 **3928 Team Neutrino** Halcyon
Ames, IA

**EPA 147**
26-16

## Tall spindexer-turret architecture; full-width linear (rack-and-pinion) intake; swerve

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|---|---|
| **SHOOTER**<br>Flywheel shooter on a 360+ degree turret atop a tall spindexer tower | **INTAKE**<br>Full-width linear intake driven by a 10-DP rack and pinion (tooth profile modified to cut with a 1/8in end mill, inspired by 3467/6328). Racks went polycarb -> 2x .2in MAXComposite + 1x .1in aluminum for tooth strength. |
| **INDEXER / HOPPER**<br>Tall spindexer feeding the turret; ~80-fuel hopper | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed (turret-aim code released) | **PUBLIC CAD / DESIGN**<br>Full CAD release (Onshape) - first comp robot fully in Onshape |

**LESSONS FOR STUDENTS**

- A rack-and-pinion intake gives a controlled straight-line deploy. The modified teeth are CNC-able.
- Test the rack under match load. The polycarb teeth sheared.
- Switching to Onshape pays off with parametric top-down design + FeatureScript despite transition hiccups.
- Plan for wear: turret wire sleeving wore quickly and became a per-event replacement.

**CAD release:** https://www.chiefdelphi.com/t/team-3928-team-neutrino-2026-cad-release/521183 · CD CAD release · Statbotics

#273 (study pick)

## 4561 TerrorBytes Cupid

Research Triangle Park, North Carolina, USA

EPA 146

46-17

### No-turret double shooter, drivetrain aims; pivot intake; twin spindexers; climber; SDS MK5n modules

**SHOOTER**

Double shooter, no turret: they did not settle on a turret design and instead align the whole robot to the hub. Aiming method: compute the field-relative angle to the hub from the x and y displacement, then subtract the robot's current heading to get a robot-relative target angle, with odometry zero facing the opposite alliance wall.

**INTAKE**

Prototyping settled on a single 3 in polycarbonate roller with WCP silicone tubing. It was enough to bring fuel in without a kicker roller or kicker plate, provided compression stayed high; at lower roller speeds the fuel would not clear the bumper. The intake pivots, and constant-force springs retract the hopper when it pivots back. The pivot gearbox bound on first assembly from tight tolerances; fixed by drilling out the motor holes and greasing it.

**INDEXER / HOPPER**

Twin spindexers, a layout the team calls a 'twindexer' after 7407's term, feeding the double shooter. Prototyping compared a spindexer against a roller floor: the spindexer needed a disruptor roller to stop fuel 'iglooing' above the main wheel and needed consistent compression; the roller floor worked best on a slight decline with zip ties spaced along the rollers so the balls did not spin in place. An early twindexer prototype peaked at 15 fuel per second and averaged about 10.3. Polycarbonate hopper, reinforced with aluminium L channel after the first plate proved floppy; a supporting 1x1 removed and nut strips replaced by zip ties to save weight.

**DRIVETRAIN**

SDS MK5n modules (the team's first season on them) in a 30 x 25 in footprint chosen to fit the double spindexer while keeping hopper volume and a wide intake. The frame uses MAXTubes to carry the double shooter with clean wire routing; a lightened aluminium bellypan stiffens it for a high-contact game. All electronics mounted inverted beneath the drivebase, as on their 2025 robot. Four-piece bumpers, which the team says puts each impact straight into the frame rather than into a gap or a 90 degree joint.

**ENDGAME**

Climber built and working. It was tested early without a drivetrain at 26 lb of load, and by the first event they were running a preload climb auto. The team notes that climbing during autonomous did not contribute to winning the auto period, but took the points anyway.

**AUTONOMOUS**

Started the first event without an auto and developed a preload climb auto on site. By Elon they were running a midfield auto consistently, which the team attributes to programming time rather than hardware.

**VISION**

Limelight with MegaTag2, fused with gyro rotation into a WPILib SwerveDrivePoseEstimator. New Limelight mounts before the Elon event removed the alignment problems seen at Wake County; the team names that as the main thing that had been holding back match performance.

**PUBLIC CAD / DESIGN**

Onshape CAD (main assembly) via the Open Alliance build thread

**LESSONS FOR STUDENTS**

- In the thread, a teammate reply walks another team through the turret-less aiming math.
- One well-chosen parameter, intake compression, deleted a subsystem.
- Driving over the bump works best diagonally, with bumpers higher than 1 in off the ground, and added drivetrain weight reduces airtime.
- The thread also records early battery drainage and brownouts, and what each fix was.
- Season result: Autonomous Award and finalist at Wake County #1 at 14-4, Impact Award and finalist at Elon, then Impact Award, the FIRST Leadership Award and a Finals win as Alliance 2's first pick at NCCMP.

**CD thread:** https://www.chiefdelphi.com/t/4561-the-terrorbytes-2026-build-thread-open-alliance/508623 · Build Thread (Open Alliance) · Statbotics 4561 · The Blue Alliance 4561

## #282 2582 PantherBots Cobra

Lufkin, TX

**EPA 144**

37-20

### Adjustable-hood dual-motor launcher; floor intake; hopper feeder; side L1 climb; swerve

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|---|---|
| **SHOOTER**<br>Adjustable-hood launcher, two flywheel motors, linear-actuator hood read by an absolute encoder; high throughput (~260 fuel/match claimed) | **INTAKE**<br>Floor intake feeding the hopper (works with hopper to empty fuel quickly) |
| **INDEXER / HOPPER**<br>Hopper + feeder delivering fuel to the dual-motor launcher | **DRIVETRAIN**<br>Swerve; drivetrain assembled early; vision still in progress late-Feb |
| **ENDGAME**<br>Side L1 climb: the CG offset from the hook needs ~0.6 friction or a dedicated gripper. | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed (integration in progress during build) | **PUBLIC CAD / DESIGN**<br>Onshape CAD + GitHub (Open Alliance blog) |

**LESSONS FOR STUDENTS**

- Watch current draw on both motors of the dual-motor launcher. A single over-current motor silently halves your shooting.
- Document openly (GitHub/Onshape/blog) to keep iteration disciplined.

**CD thread:** https://www.chiefdelphi.com/t/2582-pantherbots-build-blog-open-alliance-2026/509632 · CD build blog · Reveal 'Cobra'

## #323 1778 Chill Out HAILSTORM

Washington, USA

**EPA 134**

25-19

### Compact turret shooter fed by a spindexer; motor + single absolute-encoder turret; swerve

| | |
|---|---|
| **SHOOTER**<br>Turret-mounted launcher in a compact turret assembly. | **INTAKE**<br>Ground intake; a mid-season intake redesign delayed their practice-field time. |
| **INDEXER / HOPPER**<br>Spindexer feeding the turret. The team weighed a 'dye rotor' but rejected it as too complex; they chose the spindexer as the simplest solution with a high ceiling. The cone was finished off-CAD with cat-tongue grip-tape strips, heat-shrunk to conform for extra grip. | **DRIVETRAIN**<br>Swerve. |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Turret zeroing via a motor (relative) encoder + one absolute encoder. |
| **VISION**<br>Unconfirmed; the turret aims via encoders, and no camera is named in the thread. | **PUBLIC CAD / DESIGN**<br>Full robot CAD (Onshape) + code (GitHub, Kotlin) |

**LESSONS FOR STUDENTS**

- The absolute encoder is geared ~5:1 and supplies a re-zero value while shooting. Closest-region logic plus manual re-zero buttons cover encoder drift. The turret-zeroing scheme is a cheap alternative to a full CRT/absolute setup they could not tune in time.
- Archetype selection is risk management: choose the simplest path with a high performance ceiling.
- A recovery case: early power-consumption issues, then a hardening/tuning focus. The team rates the result among its better robots by DCMP.

**Onshape CAD:** https://cad.onshape.com/documents/c527fb2ecdd377cbb954f474/w/b89f138221c8d9223c28e242/e/c4daa58f8630b2239d8ee484 · CAD & Code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics

#359 **48 Team E.L.I.T.E.** xtremachen 29
Ohio

EPA 128
29-23-1

## Turret spindexer shooter with SOTM; roller intake; climber; swerve

| | |
|---|---|
| **SHOOTER**<br>Turret-mounted flywheel with shoot-on-the-move | **INTAKE**<br>Roller intake with a quick-change roller (a bent roller is a fast pit swap); stiffened + de-jammed after the first regional |
| **INDEXER / HOPPER**<br>Spindexer (cone geometry inspired by 4145) | **DRIVETRAIN**<br>Swerve |
| **ENDGAME**<br>Climber present but iterated for reliability (they considered removing it between events). | **AUTONOMOUS**<br>Shoot-on-the-move brought online in ~8 days between events. |
| **VISION**<br>2x Limelight 4 for AprilTag pose + shoot-on-the-move; stood up in ~8 days mid-season | **PUBLIC CAD / DESIGN**<br>Full CAD release (Onshape) - first Onshape season, 100% in-house parts |

**LESSONS FOR STUDENTS**

- Vision is a software sprint you can win late.
- Design for fast field repair, as the quick-change intake roller does. A pit swap beats a dead robot.
- Keep a written list of what to fix between events, and iterate event-to-event.
- Borrowing a subsystem detail beats reinventing it, as with the spindexer cone.

**CAD release:** https://www.chiefdelphi.com/t/frc-48-team-e-l-i-t-e-xtremachen-29-2026-cad-release/521258 · CD CAD release · Statbotics

#377 **6672 Fusion Corps** Bastion
Texas, USA

EPA 124
19-26

## Turreted flywheel shooter with kicker feed, floor intake and indexer; CTRE swerve

| | |
|---|---|
| **SHOOTER**<br>Turreted flywheel launcher. The released code exposes separate Shooter, Turret and Kicker subsystems, each driven through CTRE TalonFX controllers (Kraken/Falcon-class motors). The dedicated Turret subsystem means the launcher rotates independently of the chassis (shoot-while-aiming); a Kicker meters game pieces into the flywheel. | **INTAKE**<br>Floor intake driven by a CTRE TalonFX controller (Intake subsystem in code). |
| **INDEXER / HOPPER**<br>Indexer driven by a REV SPARK controller (NEO-class motor; model unconfirmed). Sits between the intake and the kicker/turret feed. | **DRIVETRAIN**<br>Swerve. Code uses CTRE Phoenix 6 (generated TunerConstants, TalonFX/TalonFXS module controllers, PhoenixOdometryThread) with a selectable NavX or Pigeon2 gyro. |
| **ENDGAME**<br>No climber/endgame subsystem appears in the released code. | **AUTONOMOUS**<br>PathPlanner-based autonomous routines (pathplanner deploy folder in the repo). |
| **VISION**<br>PhotonVision for AprilTag localization (VisionIOPhotonVision + photonlib vendordep in the released code). | **PUBLIC CAD / DESIGN**<br>CAD Release (Onshape) |

**LESSONS FOR STUDENTS**

- Each mechanism is split into a subsystem, a hardware IO interface and a sim implementation. Use it as a template for teams learning that architecture.
- Trace how the code is structured for shoot-on-the-move.
- The localization/auto stack is fully public and widely reusable for a mid-resource team.
- The mixed motor-controller ecosystem is a realistic example of using the right controller per mechanism.

**Onshape CAD:** https://cad.onshape.com/documents/c8f7278d84c0d231b4ae782c/w/32ba4de6819aba4f14857523/e/4f4804f7c4c66c747eefa67f · CAD + Code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics 6672 · The Blue Alliance 6672

#399 **1676 Pascack PI-oneers** Cobra Pi
Montvale, NJ
**EPA 123**
35-24-1

## Flywheel shooter; roller intake; high-capacity hopper (~80-ball usable storage); swerve

| | |
|---|---|
| **SHOOTER**<br>Flywheel shooter; continuous upgrades each event | **INTAKE**<br>Roller intake; throughput slows once the hopper is heavily loaded |
| **INDEXER / HOPPER**<br>Large hopper: ~80 balls before fuel weight bogs the intake, ~90-95 at overflow | **DRIVETRAIN**<br>YAGSL swerve, Kraken X60 drive + Kraken X44 steer, CANcoder, navX |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>3x Limelight (left/right/front), MegaTag pose estimation | **PUBLIC CAD / DESIGN**<br>CAD + code release (Onshape) |

**LESSONS FOR STUDENTS**

- Add-on Delrin 'skis' under the drivebase help the robot ride over the field bump and shield swerve motors: a cheap fix for a known hazard.
- Design to the usable capacity, not the physical overflow number.
- Aggressive event-to-event upgrades drove a steep improvement curve.

**CAD release:** https://www.chiefdelphi.com/t/1676-cad-release-2026/520302 · CD CAD + code release · Team mechanical page · Released code (practice chassis)

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#399 **3175 Knight Vision** Sisyphus
Michigan, USA (FiM)
**EPA 123**
33-24

## Shooter-and-tower assembly, four-bar intake, roller hopper floor with 3D-printed structural plates

| | |
|---|---|
| **SHOOTER**<br>The released assembly holds one subassembly named 'Shooter + Tower'. A commenter refers to the shooter side plates carrying cantilevered aluminium Limelight mounts. | **INTAKE**<br>The released assembly names the intake '4 Bar Intake'. The intake was rebuilt after their first event, at the same time as the hopper plate change. Some hopper roller types exist to clear the coaxial intake. |
| **INDEXER / HOPPER**<br>Hopper floor built on The Thrifty Bot dead-axle roller system, which the team already had in stock. The stock stub roller could not be removed without taking its whole plate off the robot, because the shaft is unthreaded into a nut. The team kept the TTB hubs and replaced the shaft with a 1/2 in hex shaft, using a shoulder screw for the front bearing, so rollers swap out in place; one side is tapped 10-32 and the other 1/4-20. That modification is not shown in the released CAD. Stock TTB rollers accept only 24-tooth pulleys, so WCP SplineXL hubs were used to run larger 3D-printed pulleys and vary the speed of alternate rollers on the hopper floor. Several roller types are used so balls do not treadmill. The hopper plates are 3D-printed structural plates, adopted after the first event for weight and rigidity; the team recommends an isogrid pocketing pattern and reports it was the most rigid pattern in their experience. | **DRIVETRAIN**<br>Unconfirmed |
| **ENDGAME**<br>No climber or endgame mechanism is named in the released assembly tree. | **AUTONOMOUS** |
| **VISION**<br>Unconfirmed. A commenter describes aluminium Limelight mounts cantilevered off the shooter plates and asks about vibration; the team does not answer and does not name a vision system. | **PUBLIC CAD / DESIGN**<br>CAD (Onshape) |

**LESSONS FOR STUDENTS**

- A COTS dead-axle roller can be made field-serviceable by changing the shaft, not the whole system.
- Check what a COTS roller constrains before committing to it, as with the pulleys on the TTB stub roller.
- Treat this release as a hardware reference to read in CAD, not a documented design.

**Onshape CAD:** https://cad.onshape.com/documents/8fa47d58e24d11fb99ea01f6/w/b35f6644b105cf745abd0a5c/e/4e87fd7cd5b50d294798ca46 · CAD release thread (CD) · CAD (Onshape) · Custom stub roller doc (Onshape) · Statbotics 3175 · The Blue Alliance 3175

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#381 (study pick)

## 2079 4-H ALARM Robotics Deadshot

Franklin, MA

**EPA 122**

26-20

### Drum shooter (no turret); compression-tuned over-bumper intake; roller indexer

| | |
|---|---|
| **SHOOTER**<br>Drum shooter; the team states it was satisfied with it. Planned to grind the hood for a higher-arching shot. SOTM not yet implemented. | **INTAKE**<br>Over-bumper intake. Its compression was too high; the team couldn't copy another team's geometry and had to run their own compression tests. Broke several times and was reinforced with aluminum plates. |
| **INDEXER / HOPPER**<br>Indexer rollers feeding the drum; planned smaller indexing rollers to raise throughput. | **DRIVETRAIN**<br>Kraken X60 swerve (TalonFXS modules) |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Shoot-on-the-move not yet implemented (planned). |
| **VISION**<br>4 PhotonVision cameras (back-left, back-right, side-left, side-right) | **PUBLIC CAD / DESIGN**<br>Onshape CAD + GitHub (Open Alliance) |

**LESSONS FOR STUDENTS**

- An intake that breaks repeatedly needs structural reinforcement, not just tuning.
- Small geometry changes affect ball flow, as with the planned smaller indexing rollers.
- Borrow hopper-capacity ideas from top teams (passive net like 5687, vertical active expansion like 1678).

**CD thread:** https://www.chiefdelphi.com/t/frc-2079-alarm-robotics-2026-rebuilt-build-thread/509762 · CD build thread · GitHub · Robot code (DeadShot)

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#496 (study pick)

## 4392 The Deceivers

Brimley, MI

**EPA 107**

24-20

### Turret shooter; star-wheel intake

| | |
|---|---|
| **SHOOTER**<br>Turret: dual Kraken X60 on a 1:1 shooter axle, two SDS brass flywheels + a single 4in Fairlane 60A wheel; adjustable hood with a 1in hood roller geared 2:1 | **INTAKE**<br>Over-bumper 4in compression wheels + star wheels (no kicker bar needed); deployed by dual mirrored rack-and-pinion (racks = 4 epoxied laminations of .120 polycarb, PETG pinions) |
| **INDEXER / HOPPER**<br>Roller hopper with a passive funnel + a kicker that flicks balls into the shooter (254-2017 style); chose roller hopper over spindexer after prototyping both | **DRIVETRAIN**<br>Swerve, Kraken X60 drive, 4.2:1, 38mm wheels, ~5.5 m/s, 0.55m track width |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>4x Limelight (one..four), MegaTag2 | **PUBLIC CAD / DESIGN**<br>Onshape (document) |

**LESSONS FOR STUDENTS**

- The rack-and-pinion intake deploy is an example of fabrication on a budget.
- They documented why they picked the roller hopper over the spindexer. Read it.
- Self-critique (3 lb of brass flywheel) models good engineering process for students.

**Onshape CAD:** https://cad.onshape.com/documents/589237fcf2364719e491ed4b · CD Open Alliance thread · Onshape CAD · Website · Robot code

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#528 **1577 Steampunk** Cerberus
Israel

EPA 107
21-20

## Turret shooter; wide intake; high-capacity storage

| | |
|---|---|
| **SHOOTER**<br>Flywheel turret shooter, ~19 BPS sustained | **INTAKE**<br>5-balls-wide intake |
| **INDEXER / HOPPER**<br>Large hopper, ~75-ball capacity | **DRIVETRAIN**<br>Swerve at an 'R2' (MAXSwerve-class) ratio; max frame perimeter, max-weight robot |
| **ENDGAME**<br>L1 climb | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>CAD + Technical Binder (Onshape) |

**LESSONS FOR STUDENTS**

- Intake width and hopper capacity are force multipliers for a high-throughput cycle machine.
- Pair student-led manufacturing with a written technical binder so the reasoning survives to next year.
- Pick a known-good swerve ratio instead of over-customizing. Spend the saved effort on scoring mechanisms.

**CAD release:** https://www.chiefdelphi.com/t/steampunk-1577-2026-cad-and-technical-binder-release/521240 · CD CAD + tech binder release · Statbotics

#710 **5811 The BONDS** Lektor
Ohio

EPA 85
15-16

## 720-deg turret shooter; 1678/27-style slapdown intake; 3D-printed spindexer; mid-season rebuild to add a hopper

| | |
|---|---|
| **SHOOTER**<br>Flywheel on a 720-degree turret rotating on a 7in X-Contact bearing; praised for tight turret wire management + ball-exclusion shielding | **INTAKE**<br>'27/1678-inspired' slapdown intake; later swapped to a rebuilt 'Buckeye' 4-bar intake mid-season |
| **INDEXER / HOPPER**<br>3D-printed spindexer; rebuilt between Wk3 and Wk5 to move the intake to the short side, freeing room to add a hopper | **DRIVETRAIN**<br>Swerve, 32x23.5in base; under-belly electronics |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>CAD release + reveal video (Onshape) |

**LESSONS FOR STUDENTS**

- A mid-season rebuild is workable. Per the team, the rebuild improved offense.
- Borrow proven geometry. The slapdown intake shows it is a fast path to a working mechanism.
- A 3D-printed spindexer is a low-cost, iterable storage option when you can't machine custom parts.

**CAD release:** https://www.chiefdelphi.com/t/team-5811-the-bonds-2026-robot-cad-release/520901 · CD CAD release · Statbotics

#730 **2183 Tigerbots**
Louisiana, USA

**EPA 83**
18-18

## 720-deg passive turret; beam-break-controlled polybelt serializer; large hopper; MK4n swerve (fit under the trench)

| | |
|---|---|
| **SHOOTER**<br>Shooter rides a central front turret with 720 deg of travel achieved via a passive mechanism (no re-zeroing). | **INTAKE**<br>Ground intake feeding the large hopper. |
| **INDEXER / HOPPER**<br>Center serializer with two polybelt roller stacks that are independently controlled (not belted together) using beam-break sensors. Rather than simply alternating, each roller is driven based on whether a ball is detected; sometimes both roll inward, changing behavior when a beam break trips. This is to de-interlock FUEL and funnel the wide hopper into one stream. (Corrected 2026-07-06 per the team CAD/Code lead.) Reached via 4+ prototypes: a spindexer (too sensitive to gravity/geometry) and a two-wheel track (balls interlocked) were rejected. V1 (top belt only) ran the Oklahoma and Magnolia regionals before the final version. | **DRIVETRAIN**<br>SDS MK4n swerve; overall height constrained to drive under the trench, which motivated the low central turret architecture. |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>Full CAD (Onshape) + subsystem build thread |

**LESSONS FOR STUDENTS**

- Documenting why prototypes failed is as instructive as the final design.
- The passive turret gets shoot-anywhere coverage without slip rings or complex wiring.
- Constraints first, mechanism second: the trench fit drove the whole architecture.

**Onshape CAD:** https://cad.onshape.com/documents/0f18097cc68763333bbf9187/w/ec4ab9bf696dddb823f66e6c/e/e6c97ba1f43ead9fa84d0197 · CD build thread (turret + serializer) · Full CAD (Onshape) · Statbotics

#844 (study pick)

## 1601 Quantum Samurai Linguist

New York, USA

EPA 75

15-11

### 360-deg turreted flywheel shooter, adjustable hood, deployable over-the-bumper intake, tower-fed hopper/indexer; swerve

| | |
|---|---|
| **SHOOTER**<br>Turret-mounted. 2x Kraken X60 belted to a 4 in OD Colson wheel via shaft at 11:18; two 1.5 lb brass flywheels belted to the same Colson shaft at 18:18 to smooth and hold rotational momentum. FUEL clearance (main-plate diameter) is 6.51 in; each flywheel retained by the side plates plus 1/8 in aluminum plates. Adjustable ABS hood driven by a Kraken X44 through an ABS double-herringbone rack and pinion at >200:1 (step-up gears), giving 21-50 deg of hood travel; anti-flex pins extend into the rack, ABS e-chain manages wiring. | **INTAKE**<br>Deployable over-the-bumper intake on the front two X2C modules. Structure: three 0.125 in polycarbonate panels joined at 90 deg with aluminum top brackets and a 1x1 0.125 in aluminum crash bar for impact. Rollers: a 2 in polycarbonate roller in silicone tubing on a 3D-printed 24T hub (WCP stub shaft) plus a 1.5 in polycarbonate roller in CatTongue grip tape on an 18T hub, belted together and to a kicker roller via a Kraken X60 at 12:18. A kicker improves FUEL compression consistency. |
| **INDEXER / HOPPER**<br>Tower-fed hopper/indexer. Indexer: two 0.25 in polycarbonate side panels 22.5 in apart with 0.125 in polycarbonate vector backing panels; two vectored 2 in intake-roller shafts plus center compliant wheels. Before reaching the tower, FUEL is agitated by 4x 1.25 in dia aluminum hotdog rollers (20.5 in long) on 20T WCP stub rollers; all shafts belted 12:24 to a Kraken X60. The tower funnels FUEL up to the turret and is reinforced with 1/4 in polycarbonate supporting plates. | **DRIVETRAIN**<br>27x27 in frame on four WCP X2C swerve modules. Double-bellypan design: two aluminum bellypans mount electronics (one for the PDP, one for the RoboRIO / Pigeon IMU / CANivore), with a bottom Slippery UHMW skid panel spanning the base for debris protection and as an anti-beaching surface. 3D-printed TPU battery sock retains the battery. |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Three Limelights for full-field localization: two mounted atop the X2C modules nearest the turret tower and a third against a hopper support stock. | **PUBLIC CAD / DESIGN**<br>CAD + Tech Binder (Chief Delphi) |

**LESSONS FOR STUDENTS**

- The turret ring is a 3D-printed 184T ring gear, driven by an aluminum pinion off a MAXPlanetary 5:1 (38.334:1). The turret is fully student-documented, so others can copy it.
- The two-stage flywheel is a way to smooth shot-to-shot energy without a heavier single rotor.
- The binder credits CAD overviews from 1323 and 1678 just before their first regional. Design reviews from veteran teams shaped quick fixes.

**CD thread:** https://www.chiefdelphi.com/t/522131 · CAD + Tech Binder release (CD) · Statbotics · The Blue Alliance

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#932 (study pick)

## 6381 Red Raider Robotics

Sheboygan, WI

EPA 70

24-23

### Simplicity-first single shooter (no turret)

| | |
|---|---|
| **SHOOTER**<br>Single fixed shooter (no second shooter, no turret). Chose one for accuracy; judged they can move balls fast enough. Turret deferred to the offseason due to inexperience. | **INTAKE**<br>Over-bumper / floor intake in its own Onshape doc; outtake variants prototyped on YouTube |
| **INDEXER / HOPPER**<br>Dedicated indexer subsystem in its own Onshape doc feeding the single shooter | **DRIVETRAIN**<br>Chassis subsystem in its own Onshape doc (swerve expected) |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Unconfirmed | **PUBLIC CAD / DESIGN**<br>Onshape - one doc per subsystem (Shooter/Indexer/Intake/Chassis/Climber) |

**LESSONS FOR STUDENTS**

- Scope to your team's experience. Skipping a turret keeps a developing team reliable rather than over-reaching.
- Fewer mechanisms, fewer failure points.
- One Onshape doc per subsystem lets students work in parallel without conflicts.
- Defer ambitious mechanisms to the offseason as a learning project rather than risking the season.

**Onshape CAD:** https://cad.onshape.com/documents/e139ac1c610559bc8767c8da · CD build thread · Onshape subsystem docs

#1109 (study pick)

## 5982 Merrill's Circuits of Destruction Doubloon

Michigan, USA

EPA 60

21-21

### WCP Competitive Concept-based scorer reworked for serviceability; quick-change intake; net hopper; swerve

| | |
|---|---|
| **SHOOTER**<br>Based on the WCP Competitive Concept shooter. | **INTAKE**<br>Quick-change intake: every roller runs on standard hex shafts, so a broken roller swaps in under a minute. Geometry revised for more consistent pickup. |
| **INDEXER / HOPPER**<br>Lightweight net hopper replacing the CC's heavy extendable hopper; lighter and more durable. | **DRIVETRAIN**<br>Swerve, built from the WCP Competitive Concept base. |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>PhotonVision for on-field targeting (not modeled in the CAD). | **PUBLIC CAD / DESIGN**<br>Full CAD (Fusion 360) |

**LESSONS FOR STUDENTS**

- Adapting a COTS concept and optimizing it for serviceability is a legitimate, time-efficient design path.
- The net hopper shows simplicity as a design win.

**Fusion 360 CAD:** https://a360.co/3NHiaZ3 · CD CAD release · Statbotics

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#1358 (study pick)

## 6333 Scavenger Robotics Marion

Massachusetts, USA (NE)

EPA 49

11-17

### Turret shooter (the team's first); spindexer (the team's first); tank (differential) drive

| | |
|---|---|
| **SHOOTER**<br>The team's first turret robot. They report reaching a consistent 3 fuel per second by the end of the season and name the spindexer size, not the shooter, as the throughput limit. | **INTAKE**<br>Unconfirmed |
| **INDEXER / HOPPER**<br>Spindexer, the team's first. They state its small size bottlenecked throughput, which held their fuel rate at about 3 per second. | **DRIVETRAIN**<br>Differential (tank) drive, not swerve. This is a correction in the project's own notes; the release post does not state the drivetrain, so confirm it in the CAD. |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>Minimal. The team states the robot did not have much going for it in software and that most functions were manually controlled. It names the lack of good vision-based automation as one of the two things that held the robot back. | **PUBLIC CAD / DESIGN**<br>Onshape CAD + technical binder |

**LESSONS FOR STUDENTS**

- The budget was for the robot, event fees, shirts and pizza together.
- The team names its own limits plainly. That is more useful to copy than a highlight reel.
- It is the only tank-drive robot among the picks. Swerve is common at the top of this list. It is not a requirement for releasing studyable CAD and documentation.
- They released CAD and a binder for a robot that did not reach DCMP or Worlds. They won their second Imagery Award at the NE WPI event.

**CAD release:** https://www.chiefdelphi.com/t/team-6333-robot-reveal-and-cad-release/518178 · Robot reveal and CAD release (CD) · Statbotics 6333 · The Blue Alliance 6333

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## #1838 135 Penn Robotics (Black Knights)

Indiana, USA

**EPA 38**
24-38

### Twin-turret twin-flywheel shooter, wide floor intake, 254-2017-style floor-roller indexer, side/center hang; swerve

| | |
|---|---|
| **SHOOTER**<br>Twin flywheel shooters on two turrets, chosen for redundancy ('one is none, two is one'; keep firing if one jams or stalls). Each turret pairs a flywheel with an adjustable hood (about 15-45 deg hood range, turret elevation about 45-90 deg); the flywheel is either direct-driven or on a 180-degree gearbox to keep the motor parallel with the flywheel shaft. A prototype used a 4in green compliant wheel (softer neoprene-tread wheels shredded the game pieces). | **INTAKE**<br>Wide floor intake; the frame was maximized on the intake/turret side for the widest possible intake. Battery and RIO were packaged under the intake to free hopper volume. |
| **INDEXER / HOPPER**<br>Indexer modeled on 254's 2017 design, with team 118 also referenced. Silicone-wrapped 1in-OD 'floor rollers' geared up to keep constant upward pull on the fuel, plus 'kickup shelf' rollers near the turrets to raise balls-per-second and act as a 'jackhammer' to clear clogs. The layout is designed to stack about 3 fuel vertically across most of the robot. | **DRIVETRAIN**<br>Swerve ('a turret/swerve' architecture); 30x25in frame. The CAD has a 'DriveTrain135-2026' tab. |
| **ENDGAME**<br>Omnidirectional hang (CAD 'Telescoping Arm' + 'Elevator'). Extending arms reach up one at a time to hang off the side or center of the ladder tower, so the robot does not block partners' side-hang spots; compliant wheels grip the bar to roll into place and reduce denting and sway. The battery is mounted under the hang to bias weight toward it. | **AUTONOMOUS**<br>The code repo includes turret azimuth math. |
| **VISION**<br>Custom machine-learning vision: their 'Southmoon' repo contains a trained model (best.pt and a best.mlpackage). The specific pipeline/product is unconfirmed. | **PUBLIC CAD / DESIGN**<br>CAD (Onshape) |

**LESSONS FOR STUDENTS**

- The twin-turret shooter is a concrete durability-vs-complexity trade for a scoring subsystem.
- The CAD is parameterized via Onshape Variable Studios. Study it for the CAD and design log, not for competition results.

**Onshape CAD:** https://cad.onshape.com/documents/20202353e5ff6b6c6fe21c64 · Open Alliance build thread (CD) · CAD (Onshape) · Code (GitHub, 2026-Rebuilt) · Statbotics 135 · The Blue Alliance 135

## #1896 157 The AZTECHS

Marlborough, MA

**EPA 36**
11-18

### Adjustable-hood turret shooter; roller intake; polycarb-roller hopper; swerve

| | |
|---|---|
| **SHOOTER**<br>Two-motor single-wheel flywheel using 4in ThriftyBot urethane wheels at ~1in compression; adjustable hood 20.0-57.5 deg | **INTAKE**<br>Roller intake; added a second Kraken X60, which sped up intaking and reduced jams |
| **INDEXER / HOPPER**<br>Polycarb-roller hopper slanted toward the uptake (grip via hockey tape / textured TPU); 254-style turret wiring; turret sensed by an effective 1:1 encoder over 360 deg | **DRIVETRAIN**<br>Thrifty Narrow Swerve via the CTRE Swerve Project Generator; uses YAMS (Yet Another Mechanism System) |
| **ENDGAME**<br>Unconfirmed | **AUTONOMOUS**<br>Unconfirmed |
| **VISION**<br>PhotonVision on two Orange Pi 5 coprocessors (load split across ~4 ThriftyCams). 3 cams on the shooter platform for localization; 1 on the turret for hub alignment. Custom AprilTag mounting plates. | **PUBLIC CAD / DESIGN**<br>Open Alliance build thread (CAD/links in-thread) |

**LESSONS FOR STUDENTS**

- Read the MJPEG-vs-YUYV trade-offs and the per-Pi hostnames for multi-camera PhotonVision.
- The second intake motor is a simple, high-impact reliability fix.

**CD thread:** https://www.chiefdelphi.com/t/frc-157-the-aztechs-2026-build-thread-open-alliance/509425 · CD build thread · Statbotics

**Sources:** Chief Delphi (reveals & Open Alliance build threads), David's Design Server / frcdesign.org (robot-pics-2026 & mechanism-examples), team CAD/code releases (Onshape, GrabCAD, GitHub), FUN "Behind the Bumpers", and Statbotics EPA & records (2026). "Unconfirmed" means a detail was not publicly published, not that it is absent.